Competing Complements in Public-Private Hospital Markets

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Abstract

Public hospitals in developing countries provide subsidized care that competes with private hospitals, while training physicians who later join the private sector. I study this tension using the staggered construction of public hospitals in Malaysia from 1996 to 2013. A new public hospital increases private hospital entry by 36 percent. This crowd-in requires both labor-supply spillovers from public hospitals and districts large enough to support private entry. Reallocating public hospitals to larger districts would raise private capacity by a further 33 percent but reduce equity in access. These findings show public hospitals trade off equity-driven allocation against total healthcare capacity.

JEL Codes: H44, I11, I18, J44, L33, O15

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1 Introduction

Whether government spending crowds out or crowds in private investment is a classic question in economics. This question is important for many markets where publicly owned providers coexist alongside private for-profit firms. The existing literature shows that offering a public option reduces private health insurance coverage (Cutler and Gruber, 1996; Gruber and Simon, 2008), affects private school enrollment (Dinerstein and Smith, 2021) and reduces demand for private investment in municipal broadband markets (Wilson, 2025). On the other hand, investing in the public sector can raise private school quality through competition (Andrabi et al., 2024), and public research labs can generate local R&D spillovers to private firms through researcher mobility (Bergeaud et al., 2025). These opposing effects reflect a tension between competition and complementarities, as public provision may reduce demand for private alternatives, but it may also generate labor spillovers or other agglomeration benefits that complement private firms.

Understanding this tension is especially important for healthcare in many developing countries, where the private sector delivers a large and growing share of healthcare services (WHO, 2020). The conventional rationale for public spending is equity driven, serving underserved areas where private markets do not enter. Yet if the public sector also produces inputs that complement private hospitals, the same public investment can simultaneously expand direct provision and crowd-in private supply. This matters for the equilibrium effect of public investment on total healthcare capacity, and importantly, how policymakers should allocate scarce public construction budgets across markets.

In this paper, I study this tension between competition and complementarities using staggered public hospital construction in Malaysia from 1996 to 2013. During this period, Malaysia built new public hospitals across 25 districts in response to public backlash against proposed healthcare privatization. I construct a new district-year panel linking administrative records on public hospital construction to private hospital entry counts from survey data. Using a staggered event study design, I find that a new public hospital crowds in an additional 0.465 private hospitals in treated districts. This main effect conceals significant heterogeneity by whether a public hospital is a specialist or non-specialist public hospital. Specialist public hospitals increase the stock of private hospitals in treated districts by approximately 0.785 or 30 percent of the pre-treatment mean. The crowd-in effect concentrates in largely populated districts, and is less pronounced in smaller districts.

Several features of the event study design support a causal interpretation. Treated and control districts follow parallel pre-treatment trends in private hospital entry, are balanced on key pre-treatment characteristics, and show no relationship between predicted private entry and public hospital construction once district fixed effects are included. The specialist result is directionally consistent across four alternative staggered difference-in-differences estimators, synthetic difference-in-differences, and matching. Additionally, because the pre-treatment trend runs in the opposite direction of the estimated effect, overturning the result would require a confounding trend that reverses sign precisely at the time of treatment (Rambachan and Roth, 2023).

Several mechanisms, such as agglomeration externalities, patient referral networks, or localized infrastructure improvements, could explain the crowd-in effects. One institutional feature exclusive to specialist hospitals is the production of specialist physicians. Specialist physicians are scarce, slow to train, and require years of postgraduate residency under attending supervision before practicing independently. In most countries, residency training is concentrated in public teaching hospitals, and graduates typically serve in the public sector under compulsory service bonds before entering private practice. Malaysia follows this pattern, with the public sector training all specialist physicians by assigning them to public specialist hospitals. After completing their compulsory service, specialists may then transition to private practice within the same district.

Three additional pieces of evidence support the labor spillovers interpretation. First, census records show that the district-level stock of private specialist physicians increases after specialist public hospital construction, with timing consistent with specialist residency duration. This occurs despite public hospitals simultaneously reducing private hospital admissions upon entry–which highlights the tension between competition and complementarities. The second piece of evidence comes from a falsification test that holds specialist public hospital training infrastructure constant and varies only whether new bed capacity was added. Between 2003 and 2012, 35 existing specialist hospitals expanded their bed capacity. The effects of these expansions on private hospital entry are null, suggesting that the crowd-in effect is not driven by capacity expansion.

The third comes from more recent data on individual-level specialist physicians. A cross-sectional snapshot in 2025 of 12,503 registered specialists shows that approximately 40 percent of specialists whose compulsory service bonds have expired practice in the private sector. The public sector trains all specialists through its monopoly on clinical residency, yet the private sector, which accounts for only 28 percent of total hospital beds, absorbs 40 percent of the specialist supply. Among 244 specialists from recent cohorts observed transitioning from public to private practice within the registry window, 62 percent remain in the same district as their last public posting or within the greater Kuala Lumpur metropolitan area. The pipeline from public training to private practice is large, geographically concentrated, and consistent with the timing and location pattern that the training-channel interpretation would predict.

The spatial pattern of private entry within treated districts further highlights the tension between competition and complementarities. Within treated districts, private entry response is non-monotonic in distance from the new public hospital: lowest in the immediate vicinity where patient competition is most intense, peaking at intermediate distances where the physician labor pool remains accessible, and declining again where commuting costs limit physician access. The non-monotonic gradient provides within-district evidence that the district-level crowd-in reflects the tension between opposing forces, rather than an artifact of cross-district heterogeneity.

A back-of-envelope calculation shows that targeting specialist hospital placement toward marginal districts with high private entry probability would crowd-in approximately 33 percent more private hospital capacity than the observed placement. This is equivalent to roughly 255 additional private hospital beds or roughly half the construction cost of a new public specialist hospital at no additional public expenditure.

If the social planner's objective is to maximize total healthcare capacity, these results suggest a type-specific allocation strategy. Non-specialist hospitals should be allocated towards districts where the private sector will not enter, while specialist hospitals can be targeted at marginal districts with high private entry probability. These results highlight an equity-efficiency tradeoff for hospital allocation policy. Allocations that maximize equity deliver minimal labor-supply spillover, while targeting districts to induce more private entry delivers less direct equity benefit. With limited fiscal space in developing countries, this tradeoff becomes particularly salient for policymakers deciding how to allocate scarce public construction resources.

Related Literature. This paper contributes to the literature on public provision and private market responses across education (Epple and Romano, 1998; Hoxby, 2000; Dinerstein and Smith, 2021), health insurance (Cutler and Gruber, 1996; Duggan and Scott Morton, 2006; Saltzman, 2023), pharmacies (Atal et al., 2024), broadband (Wilson, 2025), and consumer goods (Jiménez Hernández and Seira, 2022). Recent work shows that crowd-in is possible under specific conditions. Competitive pressures from public school funding raise private school quality (Andrabi et al., 2024), and public research labs generate local R&D spillovers to private firms partly through researcher mobility (Bergeaud et al., 2025). This paper's contribution is documenting that public provision crowds in or crowds out private investment depending on whether it expands an input that private firms also require. The evidence is consistent with a labor market spillover through public specialist training as the operative channel. While a large hospital and insurer competition literature exists for the United States (Gaynor et al., 2014; Ho and Lee, 2017; Shepard, 2022), few studies examine public-private hospital competition in developing countries, where patients pay out-of-pocket and public hospitals dominate specialist training (Das et al., 2008; Banerjee et al., 2024).

This paper also provides empirical evidence on mixed public-private competition, a literature that has been largely theoretical (Cremer et al., 1991; De Fraja and Valbonesi, 2009; Klumpp and Su, 2019) with limited empirical work on hospitals (Bloom et al., 2015; Herr, 2011). I show that competitive outcomes in mixed markets depend on the specific characteristics of public providers, particularly whether they generate complementarities that benefit private competitors. The results also connect to the place-based policies literature (Kline and Moretti, 2014; Juhász et al., 2024), showing that public provision can stimulate private investment through workforce training in addition to targeted industrial policies.

The paper proceeds as follows. Section 2 provides context and data on the Malaysian public and private hospital industry. Section 3 presents descriptive facts about the public-private hospital market. Section 4 sets out a conceptual framework for the competition-complementarity tension. Section 5 presents the empirical findings. Section 6 concludes.

2 Context and Data

2.1 History of the Public-Private Malaysian Health System

Following Malaysia's independence from the British Empire in 1957, the Ministry of Health established a network of public hospitals to ensure universal access to healthcare services. This public system operated as the dominant healthcare provider until the 1980s, when Malaysia adopted a series of broader economic liberalization policies extended to the healthcare sector. The privatization wave of the early 1980s was a key policy shift in Malaysia's health system. The government actively encouraged private investment through tax breaks for medical devices and private health insurance (Barraclough, 2000). These nationwide policies allowed private investors to enter markets based on profit considerations rather than central planning directives.

However, by the mid-1990s, political resistance to healthcare privatization emerged as a constraint on further market oriented policies. The government's initial plans to corporatize public healthcare services faced significant opposition from the ruling coalition's constituents, who viewed potential reductions in subsidized public healthcare as a threat to equitable medical care.[1] This political backlash resulted in a policy reversal that reinforced the government's commitment to maintaining a robust public healthcare system alongside the growing private sector.

The 1995 general election is an important moment that shaped Malaysia's public-private healthcare system. Rather than pursuing further privatization of public services, the government responded by expanding public hospital capacity and reaffirming subsidized public healthcare as a core public good. As a result, Malaysia developed hospital markets where private hospitals operate as an expensive alternative to a heavily subsidized public system, rather than as replacements for public provision.[2]

Importantly for the event studies, the government's renewed commitment to public hospital expansion post-1995 provides an empirical opportunity to examine the effects of public hospitals on private investment. The construction of public hospitals following the electoral mandate creates variation in public hospital entry location and timing, which I use in the event study analyses to identify causal effects on private hospital entry.

[1] In 1985, Prime Minister Mahathir Mohamad announced a series of privatization and corporatization policies across multiple industries. This led to political concern from the main coalition's constituents, as the government contemplated reducing subsidized public healthcare services. In the 1995 Malaysia general elections, the government retracted all policies related to corporatizing public healthcare services and increased the number of public hospitals to show commitment to retaining a public-dominant health system (Barraclough, 1997, 2000).

[2] The 1997–98 Asian Financial Crisis caused Malaysia's GDP to contract by 7.4 percent in 1998. Public expenditure fell short of planned targets in 1998 and 1999, likely delaying some hospital construction projects planned under the Seventh Malaysia Plan (1996–2000). Because the crisis affected all of Malaysia simultaneously, these delays are absorbed by year fixed effects in the event study specification.

2.2 Hospital Classification, Regulation, and Pricing

The Ministry of Health classifies public hospitals into four tiers based on specialization and capacity. State hospitals and major specialist hospitals (Level 4) offer the full range of specialty and subspecialty services. Minor specialist hospitals (Level 3) provide a more limited set of specialist services. Non-specialist hospitals (Levels 1–2) provide general medical and nursing care staffed by medical officers rather than specialists. In the event study analysis, I group state hospitals, major specialist hospitals, and minor specialist hospitals as "specialist" hospitals, since all three tiers operate residency training programs. Non-specialist hospitals do not train specialists.

Public hospital allocation follows a multi-tiered process embedded within Malaysia's five-year development planning cycle. Hospital funding is allocated to districts based on the Malaysia Plans, which are national development blueprints that prioritize healthcare accessibility and population coverage. The Ministry of Health collaborates with State Economic Planning Units to identify districts requiring new healthcare infrastructure based on demographic projections, existing facility capacity, and accessibility gaps (see Figure A.7 for excerpts from official planning documents). After districts receive funding allocations through this centralized planning process, local health authorities work with district officials to identify specific sites that maximize population access while considering factors such as land availability, transportation networks, and proximity to existing health facilities.

In contrast, private hospital entry operates under a regulatory framework established by the Private Healthcare Facilities and Services Act 1998. While the Ministry of Health retains approval authority for private hospital licenses, the regulatory standard primarily requires demonstration of sufficient local demand rather than adherence to national planning objectives. Private hospitals can choose any location within a district based on commercial considerations such as population density, income levels, and competitive positioning. In short, private entrants seek to make profits, while public hospitals are centrally allocated based on accessibility objectives.

The regulatory framework governing pricing differs between sectors. Public hospitals operate under a unified national pricing structure, with the government setting standardized fees for all services across the country but varying by room types. These prices are heavily subsidized. For example, a normal vaginal delivery in a public hospital costs Malaysian citizens MYR 100 (approximately USD 24) in a third-class ward while private hospitals charge a mean of MYR 3,306 for the same service. Private hospitals face a more complex regulatory environment. While the 1998 Act establishes fee schedules for physician consultations and medical procedures, it does not regulate hospital-specific charges such as room fees, meals, and ancillary services. This partial price regulation allows private hospitals significant flexibility to price their services based on local market conditions and competition.

2.3 Physician Training and the Public-to-Private Pipeline

Malaysia requires all specialist physicians to complete a four-year mandatory service obligation in public hospitals. After a physician has obtained her basic medical degree, Specialist training imposes additional years of public service. All physicians seeking specialization must complete a Master of Medicine (MMed) residency program lasting four to six years depending on the discipline, conducted in Ministry of Health hospitals.[3] Trainees receive full sponsorship under the *Hadiah Latihan*

[3]While specialist degrees are conferred by both public and private universities, clinical residency training takes place in Ministry of Health hospitals regardless of the degree-granting institution. A small number of specialists obtain qualifications abroad and register through the National Specialist Register upon return.

Persekutuan (HLP) scholarship administered by the Public Service Department, which bonds them to public service for an additional four to seven years after completing specialization.[4] After a six-month gazettement process for formal recognition, specialists may resign from government service and enter private practice once their bond obligation is fulfilled.

Specialist posting to public hospitals after MMed graduation is centrally administered by the Ministry of Health's Medical Development Division based on identified workforce needs and post availability at specific hospitals, weighted by length of prior service and prior posting in district or rural hospitals (Ismail et al., 2025). Bonded specialists under HLP cannot decline assigned posts without buying out the bond, implying that personal preference plays a limited role. When a new specialist hospital opens, MOH allocates new specialists there based on the hospital's identified workforce needs. Over time, these posts fill from successive MMed cohorts, producing a localized concentration of trained specialists.

This institutional arrangement implies that specialist public hospitals serve as both healthcare providers and training centers for the entire healthcare labor market. The combined duration of residency and bond service implies that the earliest a new specialist hospital could produce physicians available for private employment is approximately five to eight years after the hospital begins training.

During the study period (1996–2013), formal dual practice in both public and private sectors was not permitted. The Ministry of Health introduced a limited dual practice policy in 2007 that allowed public specialists to treat private patients within the same public hospital for additional fees, but this arrangement does not extend to practicing at external private facilities. The primary channel through which

[4]The bond penalty ranges from RM 500,000 to RM 700,000 depending on whether training is conducted locally or overseas. In practice, enforcement is limited: specialists who resign typically repay the penalty in small monthly installments, reducing the effective cost of early departure from public service.

publicly trained specialists enter the private sector is therefore full resignation from government service after completing bond obligations, rather than gradual moonlighting.

2.4 Data

I provide a brief overview of the data used in the event study and mechanism analyses, with further details in Appendix A. A summary of the key variables is tabulated in Table A.1. The event study data comes from a combination of administrative data and surveys conducted by the Ministry of Health.

Event Study Data. I estimate my event studies using a district-level panel across all Malaysian districts over 1996-2013.[5] The analysis focuses on 25 new public hospitals (14 specialist and 11 non-specialist) that began operations between 1996 and 2013 across 25 treated districts, using the first treatment event in each district, and their impact on the stock of private hospitals.

I construct this panel using data from the National Healthcare Establishment and Workforce Survey (NHEWS), which contains information about every hospital providing hospitalization services that was operational in 2013. Using each hospital's construction and opening dates, I backfill the count of public and private hospitals operating in each district for every year from 1996 to 2013. This implies that exits are not accounted for, though Malaysia's private hospital sector has had minimal exits during my study period (Barraclough, 2000). The final dataset includes all general and specialized hospitals providing acute curative care from both public and private sectors. I exclude specialized institutions (prison, defense, and education ministry hospitals) and long-term care facilities (rehabilitat-

[5]The sample includes districts in both Peninsular Malaysia and East Malaysia (Sabah and Sarawak). Eight of the 25 treated districts are in East Malaysia, where healthcare markets are more rural and private hospitals are less prevalent.

ive and palliative care hospitals, nursing homes, leprosy centers, and psychiatric institutions).

To understand the mechanisms driving private hospital entry patterns, I use two additional outcome data measured at the district level. Data on private specialist physicians comes from the Population and Housing Census for 1970, 1980 and 1991.[6]. I use district-level counts of self-employed physicians as a proxy for private specialist physicians, since all public specialist physicians are civil servants receiving wages rather than operating independently. This implies that my outcome is an undercount of total private specialists, but it captures the majority of private specialists who operate their own clinics or work in private hospitals.

Data on private hospital utilization comes from the National Health and Morbidity Survey conducted in 1996, 2006, and 2011. This nationally representative survey interviews approximately 59,000 respondents in 1996 and 2006, and 29,000 in 2011. The survey asks about healthcare utilization in the previous year, including private inpatient admissions. The survey weights allow for district-level estimation of private hospital utilization rates.

Data on individual specialist physicians comes from Malaysia's Annual Practicing Certificate (APC) registry. I collect the complete registry of practicing specialists and cross-match it with the National Specialist Register (NSR) to identify specialist qualification years and fields of practice. The resulting dataset covers 12,503 registered specialists observed across 2023–2026, recording each physician's primary practice location, sector (public, private, or university), specialty, medical school, and year of specialist qualification. The qualification year ranges from 1950 to 2024, providing variation in how long each specialist has had to complete bond

[6]The 2000 census adopted revised ISCO occupation codes that combine physicians, veterinarians, dentists, and other medical professionals into a single category, precluding identification of specialist physicians. I therefore cannot construct district-level physician counts during the study period itself using census data.

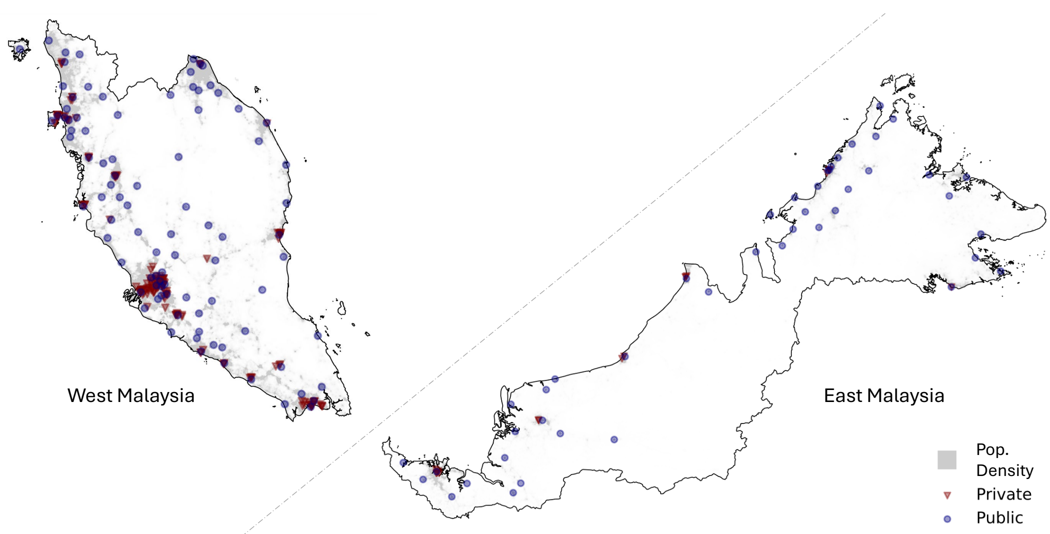
obligations and potentially transition to private practice. Because the APC is a mandatory annual license required for all practicing physicians, the data captures the near-universe of actively practicing specialists in Malaysia regardless of sector. I use this data to describe the public-to-private transition rate by qualification cohort and the geographic concentration of transitions, providing direct evidence on the physician training pipeline described in Section 2.3.

Finally, I use 'Health Facts', an annual publicly available dataset containing hospital-level information on total beds from 2004-2012, to construct an alternative treatment of hospital upgrades. Health Facts covers the same hospitals as NHEWS, allowing me to identify existing hospitals that received significant capacity upgrades (defined as increases in bed count). I observe 35 such upgrades across different hospitals during this period. This alternative treatment tests whether the private hospital entry effects are specific to entirely new public hospital construction, or also occur when existing public hospitals expand their capacity.

3 Descriptive Evidence

Private hospitals locate in urban areas while public hospitals distribute more evenly across the country. Figure 1 maps hospital locations in 2013 against population density. Figure A.3 shows how this geographic pattern has changed over time. Between 1980 and 2013, private hospitals expanded primarily in urban centers while public hospitals grew in rural areas.

Figure 1: Public and Private Hospitals Location in 2013



Note: Hospital location data are from the National Healthcare Establishment Workforce Survey (2013). Population density data are from 1km grids from the Center for Integrated Earth System Information (CIESIN).

Despite their urban concentration, private hospitals operate at a significantly smaller scale than public specialist hospitals and capture limited market share (Table A.1). The 134 private hospitals in the event study sample average 94 beds, compared to 509 beds for public specialist hospitals and 89 beds for non-specialist public hospitals. The private sector includes large corporate chains such as KPJ

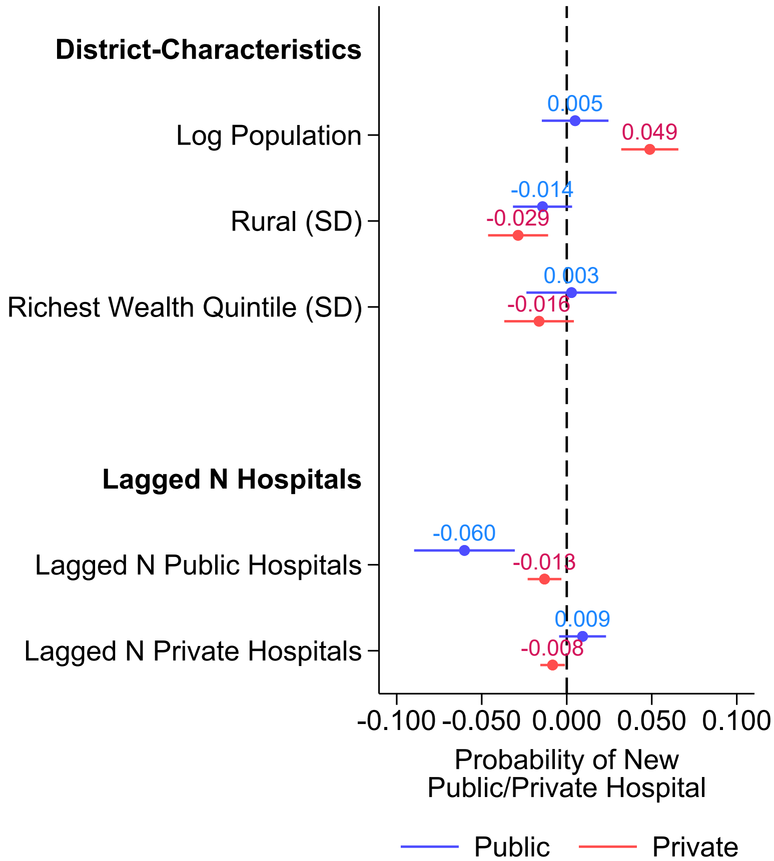
Healthcare (16 percent) and Pantai Holdings (8 percent), but 65 percent of private hospitals are independently owned, typically founded by one or a small group of specialist physicians (Table A.1, Panel A). For most private hospitals in Malaysia, recruiting specialist physicians is a key binding constraint for establishment. The law requires every private hospital to appoint a medically qualified person in charge with specialty training, and small independent hospitals typically depend on their founding physician to fill this role.

In terms of market shares, I use maternity services as an example. Private hospitals hold just 8 percent of district market share for vaginal deliveries, while public specialist and non-specialist hospitals capture 70–79 percent (Table A.1, Panel B). The remaining share reflects the outside option of home births, traditional birth attendants, and private maternity clinics. This small market share occurs despite public specialist hospitals facing congestion: public specialist hospitals have 73.9 percent bed occupancy rates compared to 47.3 percent for public non-specialist hospitals and 53.9 percent for private hospitals (Table A.1, Panel A). The congestion creates wait time dissatisfaction among public hospital patients (3.23 vs 3.82 satisfaction rating for private hospitals), yet survey respondents still rate public hospitals higher on overall quality (4.03 vs 3.83 for private hospitals; Table A.1, Panel C).

The limited overlap between public and private hospitals partly reflects segmented price markets. Private maternity services cost 3,306 MYR compared to 100 MYR for subsidized public services (Table A.1, Panel B). This price gap corresponds to differences in the patient populations they serve. Private hospital users have higher monthly incomes (2.54 vs 1.52 thousand MYR), shorter travel distances to private facilities (15.31 vs 31.82 km), and higher private insurance rates (0.52 vs 0.16; Table A.1, Panel C).

These differences extend to location decisions. Figure 2 presents the average marginal effects of district characteristics on the probability of public and private hospital entry from a logit model with year fixed effects. I omit district fixed effects to compare the characteristics of districts that received a new hospital to those that did not within the same year. The results show that the covariates predicting private entry differ from those predicting public entry. Private hospitals enter districts with higher population and lower proportion of rural residents. Public hospitals, conversely, are significantly less likely to enter districts that already have a public hospital, consistent with the Ministry of Health's stated objective of expanding access to underserved areas rather than duplicating existing public capacity. Factors that strongly predict private entry, such as population and urbanization, show weaker or statistically insignificant associations with public hospital placement. While this descriptive evidence does not rule out all potential confounding, it suggests that public hospital allocations are not primarily driven by the same profit considerations that incentivize private hospital entry.

Figure 2: Descriptive Evidence on Public and Private Hospital Entry



Note: These are average marginal effects from logit regressions of public (or private) hospital entry on a set of district characteristics with year fixed effects. The data consists of public and private entry between 1996 and 2013. The mean probability for public entry is 0.016 while it is 0.024 for private hospitals. The full coefficient plot can be found in Figure A.5. The dependent variable is a binary variable for whether a district-year receives a new public (or private) hospital. Standard errors are clustered at the district level.

4 Conceptual Framework

How do new public hospitals affect private hospital entry? A new public hospital is simultaneously a subsidized competitor, a public health infrastructure that may attract patients and providers, and a producer of the healthcare workforce. I organize these characteristics in a simple entry framework and derive the predictions that guide the empirical analysis.

Market Setup and Entry. Consider a geographically distinct healthcare market d that receives a new public hospital, characterized by two stocks: bed capacity K_g^B for patient care, and cumulative specialist training output K_g^T from residency programs. Subscript g indicates the government. Each potential private entrant $h \in \mathcal{H}$ draws a sunk cost F_h independently from a continuous distribution $G(F)$ and, upon entry, incurs a recruitment cost κ of assembling its initial healthcare team. Profit for entrant h is

$$\Pi_h \; = \; \underbrace{D_h(p_h, K_g^B)\,(p_h - c_h)}_{\text{Operating profit}} \; - \; \underbrace{F_h}_{\text{Sunk cost}} \; - \; \underbrace{\kappa}_{\text{Recruitment cost}} \; , \tag{1}$$

where $D_h(p_h, K_g^B)$ is demand for the entrant's services, decreasing in its price p_h, and c_h is variable cost. Let $\pi_h^*(K_g^B) \equiv \max_{p_h} D_h(p_h, K_g^B)\,(p_h - c_h)$ denote maximized operating profit. Entry occurs when $F_h \leq \pi_h^*(K_g^B) - \kappa$, so, normalizing the mass of potential entrants to one, the equilibrium mass of private entrants is

$$N_d \; = \; G\Big(\, \pi_h^*(K_g^B) \; - \; \kappa \, \Big) \, . \tag{2}$$

I assume that profits do not depend on the number of other private entrants to focus on the margin between the public hospital and private entry. Equation (2) shows the two channels: a new public hospital moves private entry through demand, by shifting π_h^*, or through entry costs, by shifting κ.

Potential Mechanisms. Three mechanisms are plausible. The first is demand substitution as public hospitals offer services at prices heavily subsidized relative to the private sector. Additional public bed capacity would draw patients from private hospitals, $\partial D_h / \partial K_g^B < 0$. The second is demand complementarity as patients cluster where care is available, and public facilities generate referrals and diagnoses that spill into private demand, so π_h^* increases with public capacity.

The third is input complementarity through the specialist labor market. Malaysia's public sector holds a monopoly on specialist residency training, and locally trained specialists may remain in the district after their compulsory public-service obligations expire. Cumulative training output adds to the local specialist pool,

$$L_d = \underbrace{L_0}_{\text{Existing pool}} + \underbrace{\rho\, K_g^T}_{\substack{\text{Newly trained,} \\ \text{retained}}}, \tag{3}$$

where L_0 is the pool of prior cohorts and any net inflow from outside the district, and $\rho \in [0,1]$ is the local retention share. If a deeper pool reduces the search frictions of assembling a specialist team, so that $\kappa = \kappa(L_d)$ with $\kappa'(L_d) < 0$, then training output lowers the cost of entry.

Comparative Statics. Substituting $\kappa(L_d)$ and differentiating equation (2) with respect to the two capacity stocks yields

$$\underbrace{\frac{\partial N_d}{\partial K_g^B}}_{\text{Demand channels}} = G'(\cdot) \cdot \frac{\partial \pi_h^*}{\partial D_h} \cdot \frac{\partial D_h}{\partial K_g^B}, \tag{4}$$

$$\underbrace{\frac{\partial N_d}{\partial K_g^T}}_{\text{Labor channel}} = -\,G'(\cdot) \cdot \underbrace{\kappa'(L_d)}_{(-)} \cdot \underbrace{\rho}_{(+)} > 0. \tag{5}$$

The sign of equation (4) is the net of substitution and any demand complementarity; the labor channel in equation (5) is positive but present only for training hospitals. For a hospital that expands both stocks, crowd-in occurs when

$$\underbrace{|\kappa'(L_d)|\,\rho\,\Delta K_g^T}_{\text{Recruitment-cost saving}} > \underbrace{-\frac{\partial \pi_h^*}{\partial D_h}\,\Delta D_h}_{\text{Net demand effect}} . \tag{6}$$

Market Size and the Entry Margin. Every channel in equations (4) and (5) is proportional to the density $G'(\cdot)$. Because demand scales with population, in small markets the entry threshold falls below the effective support of the sunk-cost distribution, where $G'(\cdot) \approx 0$ and all channels are muted. Whichever the mechanism, its effect concentrates in markets with marginal private entry.

Predictions. Different mechanisms would predict different aggregate outcomes. Demand mechanisms exist for both specialist and non-specialist public hospitals. Thus, both hospital types would affect private entry in the same direction. If specialist hospitals crowd-in while non-specialist hospitals crowd-out, this suggests the labor channel as the dominant mechanism.

In terms of timing effects, demand effects should be immediate, while the labor spillovers accumulate as residency cohorts complete training. If demand spillovers drive the effects, one would expect a contemporaneous effect on private entrants. On the other hand, the labor channel would produce a delayed but persistent effect.

Geography provides a third test. Patients likely substitute toward nearby care, so demand effects–substitution or complementarity–should be strongest close to the new public hospital and fade with distance. Physicians likely commute across the district, so a labor spillover would not be tied to the hospital's immediate vicinity. The two channels therefore decay at different rates relative to the new public hospital. If demand is the key mechanism, the entry response would simply attenuate with distance from the public site. If the labor channel, one would expect a decline in entry close to the new public hospital, and an increase in entry further away.

5 Reduced Form Evidence

5.1 Impact of New Public Hospitals on Private Entry

Between 1996 and 2013, Malaysia built new public hospitals across 25 districts in response to public backlash against proposed corporatization of the public healthcare system. I exploit the staggered timing of public hospital openings across 25 treated districts and 22 never-treated controls in an event study design (Figure B.1 maps these districts). District fixed effects absorb the time-invariant component of allocation, year fixed effects absorb nationwide healthcare trends, and the effect is identified from within-district changes around the timing of construction (Deshpande and Li, 2019; Cengiz et al., 2019).

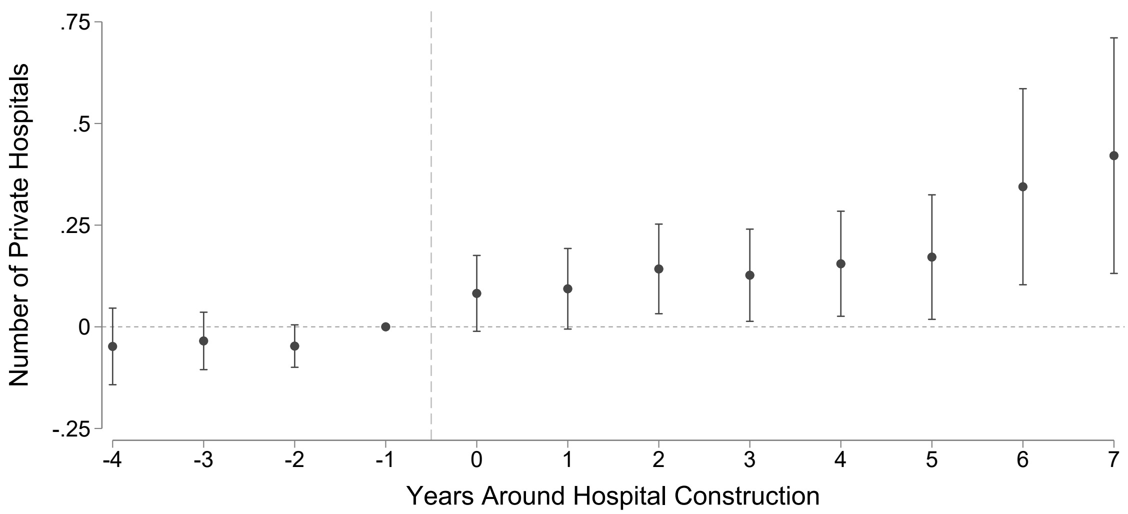
A key threat to the validity of the identification strategy is that the same factors driving private entry may also influence the location of new public hospitals. The Ministry of Health allocates public hospitals based on accessibility, population size, and congestion at existing facilities, criteria that may also drive private entry. However, balance tables between treated and control districts suggest that characteristics are largely similar across the two groups, with some differences in population, age, and rurality (Table B.1).

I estimate the interaction-weighted event study specification of Sun and Abraham (2021) (SA), which recovers cohort-specific average treatment effects on the treated by comparing each treated cohort to never-treated districts separately. Comparing treated cohorts to later-treated districts yields similar results (see Table B.9). The SA estimator avoids the contamination bias that arises in conventional two-way fixed effects estimation when treatment effects vary across cohorts (de Chaisemartin and D'Haultfœuille, 2024). For the main event study results, I use the SA estimator with robustness checks that include alternative staggered difference-in-differences estimators. Two-way fixed effects results are similar to the SA estimator and are reported in Table B.7.

$$Y_{dt} = \delta_d + \lambda_t + \sum_{e \in \mathcal{E}} \sum_{\ell \neq -1} \delta_{e,\ell} \, \mathbf{1}\{E_d = e\} \, D^{\ell}_{dt} + \varepsilon_{dt} \tag{7}$$

where Y_{dt} is the number of private hospitals in district d at time t, δ_d and λ_t are district and year fixed effects, E_d is the year district d receives its first public hospital, and $\ell = -1$ is the reference period. Three districts received more than one public hospital during 1996–2013. For these, I use the year of the first treatment, and results are robust to excluding them from the sample (Table B.8).

Figure 3: Effects of New Public Hospitals on Private Hospitals



Note: Event study estimates from Equation 7, truncated to four pre-treatment and seven post-treatment periods. Events consist of 25 newly constructed public hospitals. Each dot represents a point estimate. Vertical lines show 95 percent confidence intervals from clustered standard errors. Full coefficient table in Table B.2.

Figure 3 plots the main event study results. Coefficients are flat and close to zero in the four years preceding construction. The year of construction shows an increase in the number of private hospital relative to control districts. This effect continues to grow over the subsequent years, reflecting the dynamic response of private hospital entry following public hospital construction. By year seven, treated districts have 0.421 more private hospitals relative to control districts. The dynamic effect suggests that complementarities increase over time, though the effect at $t = 0$ also suggests that some complementarities are realized immediately.

The main effect masks significant heterogeneity by whether a public hospital is specialist or non-specialist. Figure 4 splits the event study treatment districts by hospital type. These results plot 14 specialist public hospitals and 11 non-specialist public hospitals separately. Specialist public hospitals crowd-in private entry, with coefficients rising from approximately 0.199 at the year of construction to approximately 0.861 by year seven.

In contrast, non-specialist hospitals show little to no private entry response, with coefficients hovering slightly below zero throughout the post-treatment period. The non-specialist hospital downward trend is driven by comparing treated districts with zero crowd-in with control districts that have positive private entry, which mechanically generates a negative difference. Pre-trends are not as parallel relative to the main event study. While specialist hospitals show relatively flat pre-trends, non-specialist hospitals show an almost linear pre-trend downward in the years leading up to construction. I thus interpret the non-specialist hospital results as a null effect rather than clear evidence of crowd-out.

A concern with staggered event studies based on few treatment clusters is that conventional clustered standard errors overstate statistical significance (Conley and Taber, 2011; MacKinnon and Webb, 2018; Ferman and Pinto, 2019). I implement two alternative inference procedures that do not rely on asymptotic approximations and that quantify distinct notions of uncertainty. First, a pairs cluster bootstrap that treats districts as sampled from a larger population and estimates the sampling distribution of the aggregate coefficient. Second, a randomization inference which

Table 1: Average Post-Treatment Effects on Private Hospitals

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.465		
	(0.094)		
Specialist public hospitals		0.785	
		(0.108)	
Non-specialist public hospitals			−0.171
			(0.009)
Mean Dep. Var.	1.303	1.701	0.631
Pre-Treatment Mean	1.440	2.571	0.000
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
N Districts × Year	846	648	594
R^2	0.951	0.954	0.930
Treated Districts	25	14	11
Estimator	SA	SA	SA

Notes: Each column presents results from separate regressions using the Sun and Abraham (2021) interaction-weighted estimator. Coefficients represent the average of post-treatment event study coefficients. The dependent variable is the number of private hospitals in a district. Column (1) pools all public hospitals; columns (2) and (3) separate by type. Mean Dep. Var. is the average outcome across all district-years in the estimation sample, pooling treated and control districts. Pre-Treatment Mean is the average outcome among treated districts in the pre-treatment period. Standard errors clustered at the district level in parentheses. Bootstrap and randomization inference p-values are reported in Section B.3.

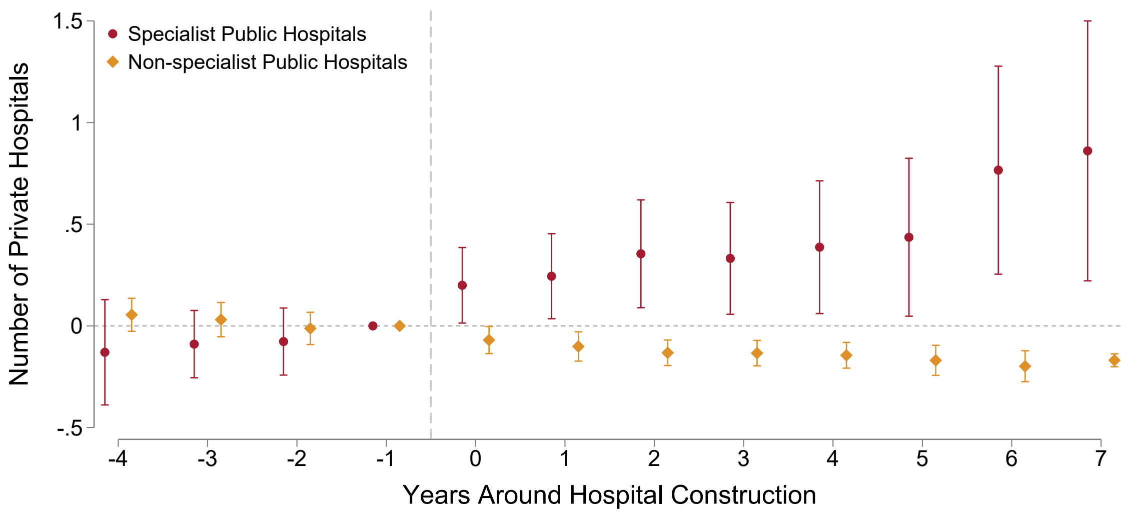
holds the districts fixed and asks how extreme the estimate is relative to the distribution generated by reassigning treatment across districts under the null of no effect (Abadie et al., 2020). Details are in Section B.3. In short, the two procedures support the conclusion that the specialist crowd-in effect is statistically significant, while the non-specialist effect is less precisely estimated.

The result is also robust across alternative specifications. Sensitivity analysis following Rambachan and Roth (2023) shows that the main result survives post-treatment trend deviations close to half of the largest pre-treatment deviation magnitude (Section B.6). The results hold under synthetic difference-in-differences

(Table B.12), coarsened exact matching (Figure B.6), exclusion of multiple-treatment districts (Table B.8), later-treated controls (Table B.9), and across four alternative staggered DiD estimators (Table B.10). Weighting by public hospital bed capacity increases the specialist effect, consistent with training capacity scaling with hospital size (Table B.11).

Overall, the main result shows that public hospital crowds in new private hospitals instead of crowding-out. The growing effect over time on private hospitals hints at the possible mechanisms. While it does not rule out mechanism that might be one-off, it does support a more persistent, gradual mechanism over time.

Figure 4: Effects of New Specialist and Non-Specialist Public Hospitals on Private Hospitals



Note: Event study estimates from Equation 7, truncated to four pre-treatment and seven post-treatment periods. Events consist of 25 newly constructed public hospitals split into specialist and non-specialist events. Each dot represents a point estimate. Vertical lines show 95 percent confidence intervals from clustered standard errors. Full coefficient table in Table B.2.

5.2 Mechanism: Healthcare Utilization and Physician Supply Spillovers

The heterogeneity above raises a question: through what channel does the specialist crowd-in occur? I test two channels directly. The demand channel–which asks whether public hospitals expand or substitute for total healthcare utilization, and the labor supply channel–which asks whether public hospitals expand the local stock of trained specialists. The demand channel uses a nationally representative household survey conducted in 1996, 2006, and 2011. The labor channel uses data from the Malaysian Population Census of 1970, 1980, and 1991, which records occupations at sufficient detail to identify specialist physicians by employment status and sector.[7] Given that neither are longitudinal with continuous time points, I use a stacked 2x2 difference-in-differences specification rather than the full event study design:

$$Y_{sdt} = \beta \cdot \text{Post}_t \times \text{Treated}_d + \alpha_{ds} + \lambda_{ts} + \varepsilon_{sdt} \tag{8}$$

where α_{ds} are district-by-stack fixed effects and λ_{ts} are year-by-stack fixed effects. Treated districts in the utilization stacks are those first receiving a public hospital of the relevant type during each stack's window (1997–2006 and 2007–2011). Districts treated during the first window are excluded from the second stack, and the comparison group in each stack comprises never-treated districts together with districts not yet treated by the stack's end year.

[7]The 2000 census adopted revised ISCO occupation codes that combine physicians, veterinarians, dentists, and other medical professionals into a single category, precluding identification of specialist physicians. I therefore cannot measure district-level physician supply during the study period itself (1996–2013) and rely on pre-period census evidence supplemented by contemporary registry data. I identify private specialist physicians by 'self-employment' category as most private specialist physicians in Malaysia are fee-for-service independent practitioners, not salaried hospital staff.

Healthcare Utilization: Treatment Substitution or Expansion? Table 2 tests whether public hospitals expand the total healthcare market or substitute for private care. Panel A shows that specialist hospitals do not affect total inpatient admissions or total outpatient visits (Columns 1 and 3), while reducing private inpatient admissions by nearly half relative to the pre-treatment baseline (Column 2) and private outpatient visits by more than a third (Column 4).

This result, in which public utilization grows at private utilization's expense without expanding total utilization, is consistent with patient substitution toward the new public option. Panel B shows that non-specialist hospitals produce directionally similar but imprecise reductions across all utilization measures.

The substitution result partially rules out the demand channel as an explanation for the specialist crowd-in: specialist hospitals draw patients away from private facilities rather than expanding demand for them. Two additional facts reinforce this. Cross-sector referrals are negligible: private-to-public referrals account for only 0.05 percent of public hospital admissions in 2013 public electronic health records. In addition, specialist hospitals do not increase total outpatient volume (Column 3), ruling out a congestion channel that would occur through overflow from saturated primary care. While there might be spillovers in other dimensions of healthcare utilization, the evidence here suggests that the demand channel is not the primary driver of the specialist crowd-in effect.

Table 2: Effects of Public Hospitals on Healthcare Utilization

	Inpatient Admissions (in 10,000)		**Outpatient Visits (in 10,000)**	
	Total (1)	Private (2)	Total (3)	Private (4)
Panel A. Specialist Public Hospitals				
Post × Treated	−0.167	−0.299	0.086	−1.371
	(0.330)	(0.140)	(0.659)	(0.671)
Mean Dep. Var.	1.409	0.428	2.008	1.630
Pre-Treatment Mean	2.610	0.651	4.333	3.703
Observations	120	120	120	120
R^2	0.871	0.815	0.865	0.933
Panel B. Non-Specialist Public Hospitals				
Post × Treated	−0.369	−0.111	−0.515	−0.304
	(0.228)	(0.099)	(1.027)	(0.231)
Mean Dep. Var.	0.842	0.293	1.511	0.880
Pre-Treatment Mean	0.465	0.022	2.705	0.490
Observations	114	114	114	114
R^2	0.834	0.804	0.750	0.942

Notes: Stacked difference-in-differences estimates using equation (8). Each column reports the average treatment effect of public hospital construction on healthcare utilization from the National Health and Morbidity Survey (1996, 2006, 2011). Panel A: specialist public hospitals (14 treated districts). Panel B: non-specialist public hospitals (11 treated districts). Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Standard errors clustered by district in parentheses.

Private Specialist Physician Supply. The utilization presents a puzzle: specialist hospitals reduce private admissions by nearly half and private outpatient visits by more than a third, yet private hospitals still enter. If competition on the demand side is strong, the supply side must offer a countervailing force. Table 3 provides direct evidence for the specialist physician training channel, measuring the stock of specialist physicians at the district level using census data at three-year lags from hospital construction.[8]

Specialist hospitals more than double the total physician stock in treated districts (Column 1) and nearly triple the stock of private specialist physicians (Column 3), raising the private specialist share of total physicians by approximately 28 percentage points from a pre-treatment mean of 38 percent (Column 5). The lag structure traces the training pipeline directly: at lag 0, the immediate effect on the private specialist stock is approximately 55 physicians, growing to approximately 108 by lag 3 as training cohorts graduate and transition to private practice (Table B.4). Non-specialist hospitals, which are staffed primarily by nurses and general practitioners, show no effect on any physician outcome (Columns 2, 4, 6).

[8]The three-year lag from hospital construction to census observation captures both immediate physician recruitment (staff hired when the hospital opens) and early training cohort graduates. This is a different time horizon from the 4–7 year compulsory service bond described below, which measures time from specialist qualification to private practice transition. Table B.4 shows effects at lags 0 through 5; the stock grows monotonically as successive cohorts complete training.

Table 3: Effects of Public Hospitals on Specialist Physician Supply (3 Years Post-Construction)

	Total Physicians (in 1,000s)		**Self-Employed Specialists (in 1,000s)**		**Self-Employed Share of Total**	
	(1)	(2)	(3)	(4)	(5)	(6)
Specialist	0.192		0.108		0.275	
	(0.117)		(0.047)		(0.135)	
Non-specialist		−0.024		−0.008		−0.022
		(0.027)		(0.006)		(0.015)
Mean Dep. Var.	0.115	0.054	0.041	0.014	0.218	0.093
Pre-Treatment Mean	0.150	0.000	0.057	0.000	0.375	0.000
District × Stack FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × Stack FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	62	66	62	66	62	66
R^2	0.863	0.923	0.858	0.880	0.940	0.985

Notes: Stacked difference-in-differences estimates using census data (1970, 1980, 1991). Effects measured 3 years post-construction to allow specialist training programs to mature; lag robustness is in Table B.4. Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Non-specialist treated districts have zero pre-treatment physician counts because these hospitals are constructed in areas without existing medical infrastructure. Standard errors clustered by district in parentheses.

Descriptive Evidence from Individual Physician Registry Data. The census evidence measures the stock of physicians but cannot directly observe the flow of specialists from public training to private practice. I provide direct individual-level evidence on this flow using Malaysia's Annual Practicing Certificate (APC) registry, a mandatory annual license covering 12,503 registered specialists observed from 2023 through 2026.[9]

Figure 5 presents two facts supporting the mechanism. Panel A shows the share of specialists in private practice by qualification cohort. Among recent cohorts (2022–2024), fewer than 10 percent practice privately, consistent with the 4–7 year compulsory service bond following government-sponsored training.[10] As bonds expire for earlier cohorts, the private practice share rises steeply, reaching approximately 40 percent for cohorts qualifying before 2015. This implies a long-run steady state in which approximately two in five specialists eventually transition to private practice.

Panel B shows the geographic concentration of public-to-private transitions among 244 specialists from 2015–2019 cohorts observed making this move within 2023 through 2026. One-third remain in the same district as their last public posting, and an additional 29 percent remain within the greater Klang Valley metropolitan area which encompasses the capital city of Kuala Lumpur and multiple smaller districts.[11] Together, approximately 62 percent of movers remain in the same local

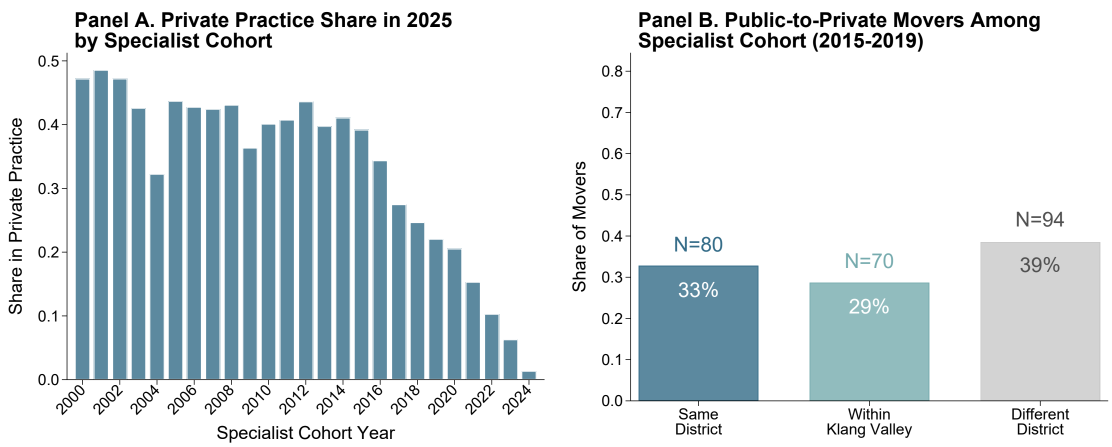
[9]Specialist qualification year refers to the year in which the specialist degree (e.g., Master of Medicine) is conferred by the training institution. The APC data records each physician's primary practice location, sector (public, private, university, military), and any secondary practice locations for each year of active licensure.

[10]Under the Medical Act 1971 and Medical Regulations 2017, specialist training through the Master of Medicine program is funded by government scholarship (*Jabatan Perkhidmatan Awam*), which carries a compulsory 4–7 year service bond in public hospitals following a 6–18 month gazettement period.

[11]The Klang Valley encompasses Kuala Lumpur, Petaling, Hulu Langat, Gombak, Klang, Putrajaya, and Kuala Langat.

labor market, consistent with specialist hospitals expanding the local supply of trained specialists who subsequently establish or join nearby private practices.

Figure 5: Direct Evidence on the Public-to-Private Specialist Physician Pipeline



Note: Panel A shows the share of specialists currently in private practice by year of specialist qualification, using 12,503 registered specialists from Malaysia's Annual Practicing Certificate (APC) registry (2025). Panel B classifies 244 public-to-private transitions among specialists from 2015–2019 qualification cohorts by geographic proximity to their last public hospital posting.

The census evidence in Table 3 covers 1970–1991, predating the main study period of 1996–2013. Two institutional facts suggest that the training channel is valid throughout the study period. First, the legal framework has been the same since 1971. The Medical Act 1971 establishes compulsory public service and annual practicing certificate requirements that have remained in place, and the government scholarship that funds specialist training with a compulsory service bond has operated since then[12]. Second, the APC registry shows the pipeline remains active today, with approximately 40 percent of specialists eventually transitioning to private practice.

[12]The first significant disruption being the introduction of contract employment in 2017, after the study period (Ismail et al., 2025). The only major regulatory amendment, in 2012, introduced a formal National Specialist Register. Before this, no specialist registration system existed, and transitions from public to private practice required only an Annual Practicing Certificate. The pipeline was therefore less regulated during the study period than it is today.

Hospital Upgrade Falsification Test. Between 2003 and 2012, 35 public hospitals underwent significant bed capacity expansions while maintaining their status as specialist or non-specialist hospitals. Table B.14 shows that specialist hospital upgrades have limited effects on private entry. This is consistent with the idea that it is the specialist training infrastructure, rather than general capacity expansions, that drives the crowd-in effect. Further details are in Section B.11.

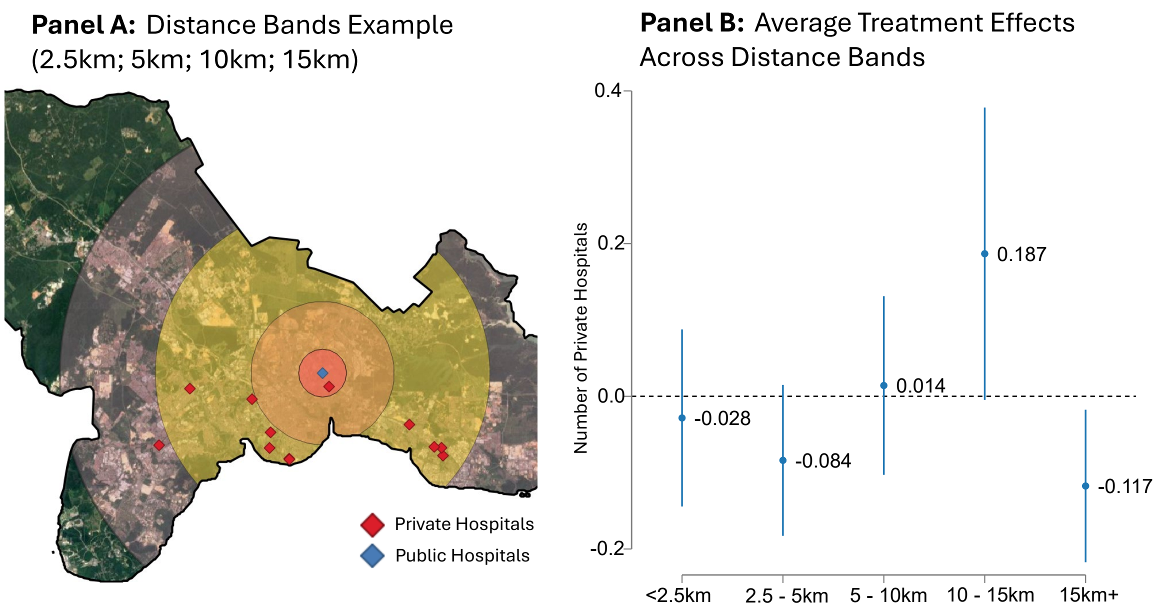
5.3 Within-District Spatial Evidence: Competition versus Complementarity

The mechanism above shows both competition and complementarity channels occur at the district level. But what about within districts? I use a Hotelling model intuition to show the tension between competition and complementarities in how private entrants locate. Specifically, I exploit within-district variation in the distance between new specialist public hospitals and private entrants.

In theory, competition would incentivize private entrants to locate far from the new public hospital. Physician supply complementarity would show a different pattern as physicians may reside all over the district. Thus, one would expect the private entry response to be non-monotonic in distance: entry should be lowest in the immediate vicinity where competition dominates, highest at an intermediate distance where the labor pool remains accessible but competition is attenuated, and falling again at greater distances where commuting costs limit physician access.

To capture this spatial gradient, I estimate Equation 7 separately for the number of private hospitals within five mutually exclusive distance bands from the new specialist public hospital: 0–2.5km, 2.5–5km, 5–10km, 10–15km, and beyond 15km. Never-treated districts serve as the comparison group, with the dependent variable counting all private hospitals across the entire district.

Figure 6: Effects of Specialist Public Hospital on Private Entry by Distance



Notes: Panel A maps example distance bands for the Johor Bahru district. The rings represent 2.5km, 5km, 10km and 15km from the newly constructed public hospital. Districts with multiple specialist public hospitals are omitted. Panel B plots post-treatment estimates from Equation 7 across five distance bands. The dependent variable in each regression is the number of private hospitals within the specified distance band in treated districts; the comparison group counts all private hospitals across the entire district in never-treated districts. Pre-treatment mean number of private hospitals: 0.29, 0.29, 1.43, 0.21, and 0.36 for bands 1 through 5. Standard errors clustered at the district level.

The spatial pattern in Figure 6 Panel B traces the predicted non-monotonic gradient across the five distance bands. Private entry is negative in the immediate 0-5km vicinity, with point estimates of approximately -0.028 in the 0-2.5km band and approximately -0.084 in the 2.5-5km band, consistent with patient competition being most intense at short distances.

The 5-10km band, which contains the highest pre-treatment private hospital density (a pre-treatment mean of 1.43 versus under 0.4 for the other bands), shows essentially zero effect. Private entry then peaks at approximately 0.19 in the 10-15km band, where private hospitals can presumably access the shared physician labor pool while avoiding direct patient competition. Beyond 15km, the effect turns negative again at approximately -0.12, consistent with commuting costs limiting physician access at greater distances.

5.4 Heterogeneous Effects by Baseline District Characteristics

Where does the crowd-in effect occur across treated districts? The conceptual framework shows that a specialist hospital lowers private entry costs through the physician channel, but entry occurs only where private demand is available. Crowd-in should therefore concentrate in districts near the private entry threshold and with access to a physician labor pool, and should be absent where demand is too thin for private entry.

I test this prediction by interacting my treatment variable with four separate pre-treatment characteristics that capture distinct channels. First, a binary indicator for a private hospital in 1996 measures viability–the market has already proven it can sustain private entry. Log population in 1996 measures latent demand and how close a market sits to the entry threshold. The log stock of public-sector doctors measures the depth of the local physician pool that a private entrant can recruit from. A binary indicator for a public hospital in 1996 captures whether private hospitals simply cluster around public health infrastructure regardless of whether it trains physicians.

I estimate two-way fixed-effects specifications on the full panel of districts between 1996 and 2013:

$$Y_{dt} = \delta_d + \lambda_t + \beta_1 \,\text{Spec}_{dt} + \beta_2 \,(\text{Spec}_{dt} \times X_d) + \gamma \,\text{NonSpec}_{dt} + \varepsilon_{dt} \tag{9}$$

where Spec_{dt} and NonSpec_{dt} indicate post-treatment years for specialist and non-specialist construction, and X_d is one predetermined characteristic, centered at the sample mean for the continuous measures. β_1 measures the specialist effect for a district at the mean. Column 5 includes all four interactions jointly. The non-specialist indicator enters as a control in every column so that non-specialist-treated districts do not serve as untreated comparisons during their own post-treatment years.

Table 4 shows that every characteristic predicts a larger specialist effect in isolation (Columns 1–4). However, when stacking all four interactions to separate these channels, log of population explains the crowd-in (Column 5). The log population interaction, at approximately 1.307, is essentially unchanged. The doctor interaction falls to approximately 0.162 but is statistically significant. This is consistent with a residual labor-market margin beyond what population absorbs. The public hospital interaction collapses from approximately 1.834 to essentially zero. The crowd-in therefore scales with district population.

Table 4: Heterogeneity of Specialist Hospital Effect by Pre-treatment District Characteristics

	Number of Private Hospitals				
	(1)	(2)	(3)	(4)	(5)
Specialist public hospitals	−0.189	−0.309	0.311	0.037	−0.265
	(0.094)	(0.245)	(0.236)	(0.200)	(0.254)
× Private Hospital in 1996	1.662				−0.564
	(0.547)				(0.490)
× Log Population (1996, centered)		1.287			1.307
		(0.266)			(0.280)
× Log Public Doctors (1991, centered)			0.354		0.162
			(0.121)		(0.084)
× Public Hospital in 1996				1.834	0.148
				(0.690)	(0.459)
Non-specialist public hospitals	−0.251	−0.251	−0.251	−0.253	−0.251
	(0.070)	(0.070)	(0.070)	(0.070)	(0.070)
Mean Dep. Var.	0.738	0.738	0.738	0.738	0.738
District Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
N Districts × Year	2,340	2,340	2,340	2,340	2,340
R^2	0.954	0.957	0.955	0.955	0.958

Notes: Each column reports a TWFE regression of the number of private hospitals on separate post-treatment indicators for specialist and non-specialist construction, with the specialist indicator interacted with one predetermined characteristic; Column (5) includes all four interactions. The sample excludes three districts with missing 1991 doctor counts. Standard errors clustered at the district level in parentheses.

5.5 Private Capacity Gains From Public Hospital Reallocation

The heterogeneity in Table 4 shows that the specialist crowd-in effect scales with district size. This section provides a back-of-envelope calculation that quantifies what this implies for the placement of new public hospitals. Details are in Section B.13.

Using the log population specification (Column 2 of Table 4), I predict the number of private hospitals crowd-in under the observed placement of the 14 specialist hospitals. Summed over the 14 specialist-treated districts at 2013, the status quo crowds in approximately 14.0 additional private hospitals. Six of the 14 specialist hospitals were placed in districts where the implied crowd-in effect is near zero.

Now, consider an alternative placement of all 14 specialist hospitals. Ranking the 84 candidate districts (those without an in-window specialist treatment or pre-1996 specialist hospital) by predicted private hospital count from a census-panel regression, I replace these six placements with the top six candidates. This reallocation would have crowded-in approximately 4.6 additional private hospitals net, adding approximately 5.1 at the new sites while forgoing approximately 0.4 at the observed districts, a 33 percent expansion of crowded-in capacity from the same number of specialist hospital openings (Table B.16).

Translating to bed capacity using the 1996–2013 entrant mean of 55 beds per private hospital, the alternative placement would have crowded-in approximately 255 additional private beds beyond the 770 crowd-in under observed placement. Valued at a mid-period nominal construction cost of approximately RM 300,000 per public bed, the 255 additional private beds are equivalent to roughly RM 76 million (18.6 million USD)of public hospital construction, or about half the construction cost of one new public specialist hospital.

However, crowded-in private beds are not perfect substitutes for public beds. Private hospitals charge prices that exclude lower-income patients, and the service mix differs across sectors. The reallocation therefore raises total capacity at the cost of moving public capacity away from districts that are potentially most in need of such services.

6 Conclusion

This paper shows that public provision can crowd-in private investment when public providers produce inputs that complement private firms. In Malaysia's hospital market, specialist public hospitals train specialist physicians that crowd-in private entrants, while non-specialist public hospitals that lack these programs do not. Using hospital surveys, administrative data, census, physician registry, and within-district spatial evidence, I show that the crowd-in occurs in populated districts where private demand can support new entry, and occurs through expanded local specialist physician supply.

The literature on public-private market interactions has documented a variety of equilibrium responses, but most of this work has the public sector acting on the output market, as a substitute, competitor, or quality benchmark (Cutler and Gruber, 1996; Atal et al., 2024; Andrabi et al., 2024; Wilson, 2025). The mechanism in this paper acts on the input side. The public sector holds a monopoly on the production of a complementary input that private firms require for entry. The closest precedent is Bergeaud et al. (2025), in which researcher mobility from public labs generates a knowledge spillover to private R&D, though the spillover there flows through ideas embedded in workers rather than through a public sector labor monopoly.

This labor complementarity is not unique to hospitals. Public flight academies produce aviators who later enter commercial aviation; public legal aid offices train litigators who move into private practice; public school systems develop the teachers whom private schools hire while competing for their students. In each case, public provision competes with private firms for demand while providing labor supply, and the net effect on private investment depends on which force dominates. The condition that gives the training channel its magnitude is a public monopoly or near-monopoly on training, which leaves private firms unable to substitute away from the publicly trained workforce.

These findings reveal an equity-efficiency tradeoff for allocating scarce public hospital construction budgets in low- and middle-income countries. A type-specific allocation rule that targets specialist hospitals toward districts with high private entry probability and non-specialist toward underserved areas would have crowded-in more private hospital capacity than the observed allocation, at no additional public construction cost. The crowded-in capacity is not free of distributional consequences, however. Private hospitals enter only where willingness-to-pay supports profitable entry, which excludes populations who cannot afford private prices. This implies that the allocation that maximizes total beds concentrates the induced capacity in markets that are already partially served. If the planner's objective is purely to expand access in underserved areas, the observed equity-driven allocation is defensible. If total capacity also matters, the type-specific rule is justified.

This paper takes a positive, not normative stance. The main result identifies a channel through which public provision generates private spillovers, without claiming the resulting crowd-in is welfare enhancing. Two gaps stand between this and a full welfare assessment. First, I do not observe healthcare quality, so if physician migration to private practice degrades public hospitals, the gains from private expansion may come at the expense of populations who rely on public facilities. Second, I document the labor-supply spillover without modeling the physician labor market itself. Understanding how bond length, training capacity, and geographic restrictions jointly determine the spillover's magnitude would clarify when public provision stimulates complementary private investment and when the physician outflow undermines the public system.

When the public sector holds a monopoly on inputs that private firms require, expanding public provision can stimulate private investment. The channel documented here operates wherever specialists must train in the public sector, an institutional feature common across many countries.

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Appendix: Table of Contents

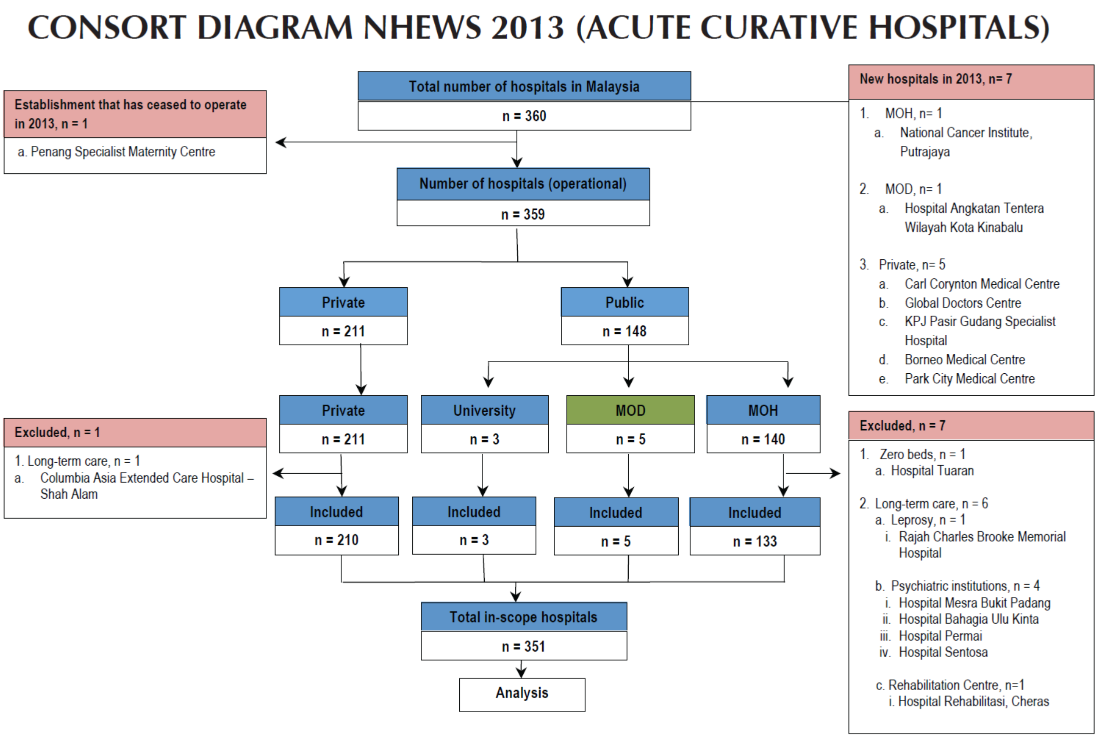
A Details on Data and Context

A.1 Hospital Survey Data Details

Hospital survey data comes from the National Healthcare Establishment and Workforce Survey (NHEWS) (2010-2013) conducted by the Clinical Research Centre. This survey provides me with a panel dataset of hospitals, whether the hospital provides certain services, the year in which hospitals began providing services, and year-specific levels of admission, congestion.

The survey gathers information on hospitals in the country regarding their services, with emphasis on specialized clinical services, facilities, medical devices, and health workforce. The NHEWS survey covers all acute curative hospitals and related specialty services for both public and private sectors. This survey asks all facilities that provide inpatient admissions in Malaysia. The survey respondent is the person-in-charge for the administrative department of hospitals. Response rates for public hospitals are 100 percent but for private hospitals it is 83.6 percent for those with less than 20 medical subspecialties and close to 90 percent for those with more than 20 medical subspecialties.

Figure A.1: Sample Details for NHEWS (2010-2013)



Notes: This figure presents sample details from the National Healthcare Establishment and Workforce Survey (NHEWS) in 2013.

A.2 Additional Survey Data Details

The National Health and Morbidity Survey is a nationally representative, two-stage (states and urban-rural status) stratified randomly sampled household survey in Malaysia. The survey asks respondents on hypothetical choice scenarios for birth deliveries after sociodemographic questions–which allows me to observe quality ratings and waiting time ratings. Specifically, the question asks:

"Which is the main health facility you would go to in the following situations?" "For birth delivery, where would you go?".

The respondent could choose exactly one from four possible responses: "government", "private", "traditional/complementary/alternative health facility" and "will not go to any facility". Following the hypothetical choice questions, the survey asks individuals on their perception of quality. The survey question asks the respondent on ratings on a 1 to 5 Likert scale, which encompasses 12 different questions of quality for outpatient care and inpatient care, in both the public and the private sector. I specifically use the questions *'Based on your perception or impression, how would you rate the government and private hospital on the following aspect ...'*. First, *'The waiting time to see a doctor once arrived at a hospital'* followed by *'Your overall impression'* Answers do not correspond to specific types of health conditions, and instead refers generally to the public and private sector.

Figure A.2: Quality Perception Survey Questionnaire

Sangat tidak bagus *Very Poor*	Tidak bagus *Poor*	Sederhana *Fair*	Bagus *Good*	Sangat bagus *Excellent*
1	2	3	4	5

(-7) TT (-9) EJ

Berdasarkan tanggapan atau kepercayaan anda [kepada penemuramah: sekiranya responden menghadapi masalah, anda boleh bantu dengan mencadangkan, e.g. Daripada perkhabaran/pengalaman keluarga, rakan-rakan anda, pengalaman anda sendiri], bagaimana anda menilai **HOSPITAL** kerajaan dan swasta (pesakit dalam) pada aspek berikut ... / *Based on your perception or impression [to interviewer: if respondent has trouble answering, you can help them by suggesting, e.g. What you hear from your relatives and friends, other's experience, own experience], how would you rate the government and private **HOSPITAL** (inpatient) on the following aspect …*

Tanya semua soalan berkenaan fasiliti Kerajaan dahulu, diikuti dengan Swasta.

		Hospital / *Hospital*	
		Kerajaan *Government*	Swasta *Private*
AC215	Kesesuaian lokasi hospital *Convenience of hospital location*		
AC216	Boleh memohon bilik persendirian / tidak berkongsi dengan ramai pesakit lain / *Ability to ask for a private room / sharing with less people*		
AC217	Keselesaan hospital (cth: kebersihan, susun atur kerusi, ruang, dll.) / *Comfort of hospital (e.g. cleanliness, setting of chairs, space, etc.)*		
AC218	Adanya ujian/ peralatan perubatan *Availability of investigations/ medical equipment*		
AC219	Adanya doktor pakar di hospital pakar *Availability of specialist (s) at the specialist hospital*		
AC220	Dibenarkan memilih doktor *Allowed to choose the doctor*		
AC221	Tempoh menunggu untuk berjumpa doktor sebaik tiba di hospital *The waiting time to see a doctor once arrived at the hospital*		
AC222	Masa yang diluangkan oleh doktor untuk pesakit *The amount of time the doctor spends with a patient*		
AC223	Kebolehan doktor memberi diagnosis dan memberi rawatan yang betul / *The ability of the doctor to give you the correct diagnosis and treatment*		
AC224	Kejelasan penerangan doktor berkenaan penyakit, ujian dan prosedur / *Clarity of doctor's explanation regarding the illness, test and procedure*		
AC225	Budi bahasa dan kesediaan doktor, penolong pegawai perubatan & jururawat untuk membantu / *Courtesy and helpfulness of doctor, assistant medical officer and nurse*		
AC226	Keberkesanan perkhidmatan / rawatan *The outcome of services / treatment*		
AC227	Caj rawatan *Treatment charges*		
AC228	Pandangan anda secara keseluruhan *Your overall impression*		

10 Borang Soal Selidik Individu Responden 13 Tahun Dan Ke Atas / *Individual Questionnaire Respondent 13 Years Old and Above*

Note: This figure shows the questions asked in the National Health and Morbidity Survey (NHMS) on quality perception for hospital services.

A.3 Additional Tables and Figures on Data and Context

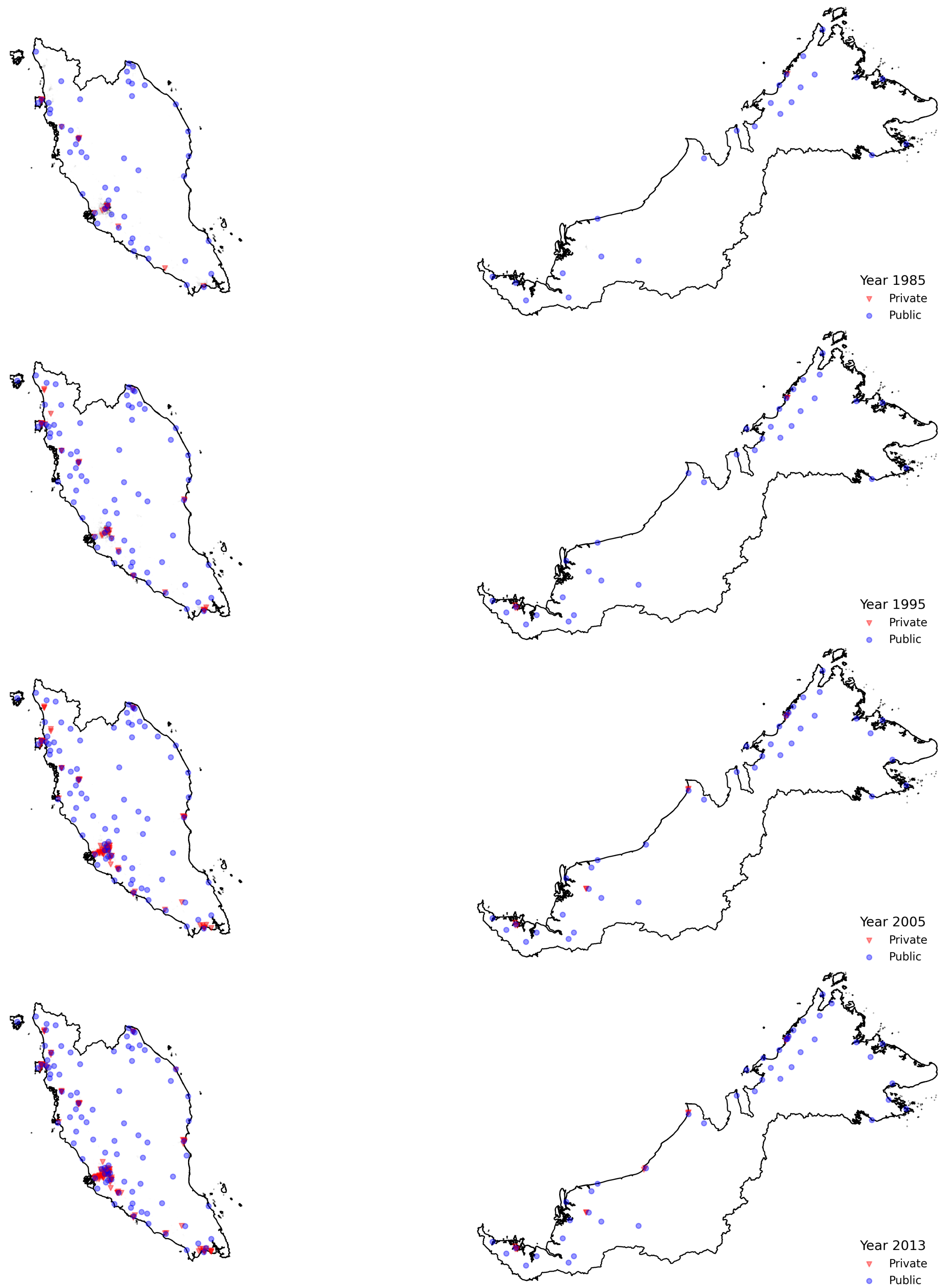
Table A.1: Summary Statistics

	Public Hospitals Specialist	Public Hospitals Non-Specialist	Private Hospitals
A. Hospital Characteristics (2013)			
N Hospitals	61	74	134
Avg. Physician	226.39	12.36	34.89
Avg. Other Staff	719.79	80.91	133.29
Avg. Beds	509.08	88.80	94.06
Avg. Inpatient Admissions	34,291	5,432	8,395
Avg. Outpatient Visits	96,276	52,539	28,116
Avg. Bed Occupancy Rate (%)	73.9	47.33	53.88
Ownership Group (%)			
Government	61 (100%)	74 (100%)	-
Independent	-	-	87 (65%)
Columbia Asia	-	-	11 (8%)
KPJ	-	-	22 (16%)
Pantai	-	-	11 (8%)
Sime Darby	-	-	3 (2%)
B. Maternity Services			
Avg. Vaginal Deliveries	3,176	586	589
Avg. District Market Share (Deliveries)	0.70	0.79	0.08
Price (MYR)	100	100	3,306

	Survey Data: Public Hospitals	Survey Data: Private Hospitals
C. Survey Data		
Indv. Monthly Income (MYR)	1.52	2.54
Distance Public (km)	13.17	10.28
Distance Private (km)	31.82	15.31
Private Insurance	0.16	0.52
Chronic Disease	0.70	0.61
Quality Rating (1-5)	4.03	3.83
Wait Time Satisfaction	3.23	3.82

Notes: Panel A shows characteristics for 269 hospitals from the National Healthcare Establishment and Workforce Survey (NHEWS). Panel B presents maternity service statistics for normal vaginal deliveries from Ministry of Health electronic health records (SMRP for public, PHDD for private hospitals). Public hospital prices reflect standardized subsidized rates for third-class wards. Private hospital prices are minimum advertised rates from primary data collection (websites, social media, direct contact). Panel C shows stated preferences from 15,296 families with childbearing intentions in the National Health and Morbidity Survey (NHMS) 2015, split by hospital type preference. Distance measured as straight-line distance from households to nearest hospital within each district. Income in thousands of MYR. Quality rating and wait time satisfaction on 1-5 Likert scales.

Figure A.3: Public and Private Hospital Locations (1982-2013)

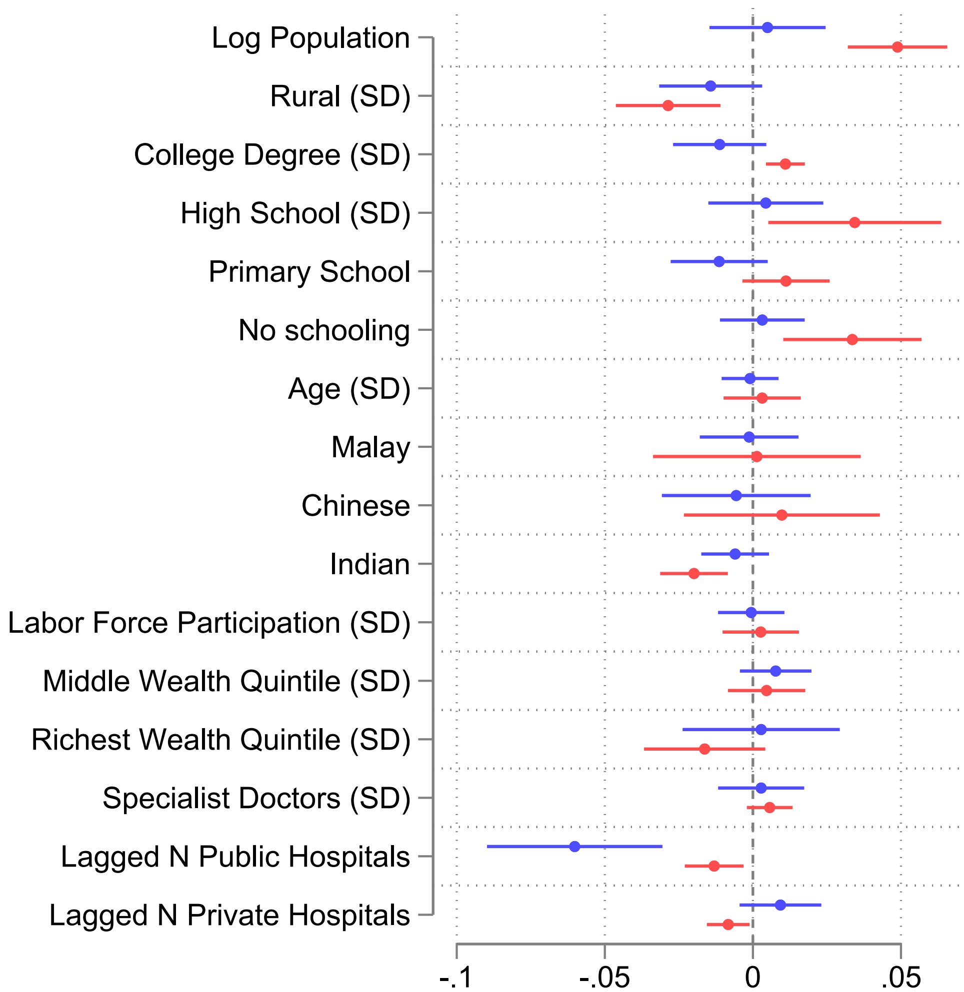


Note: Data on hospital locations are from the National Healthcare Establishment Workforce Survey (2013). This figure shows the locations of public and private hospitals in Malaysia from 1982 to 2013.

Figure A.4: Example Hospital Images

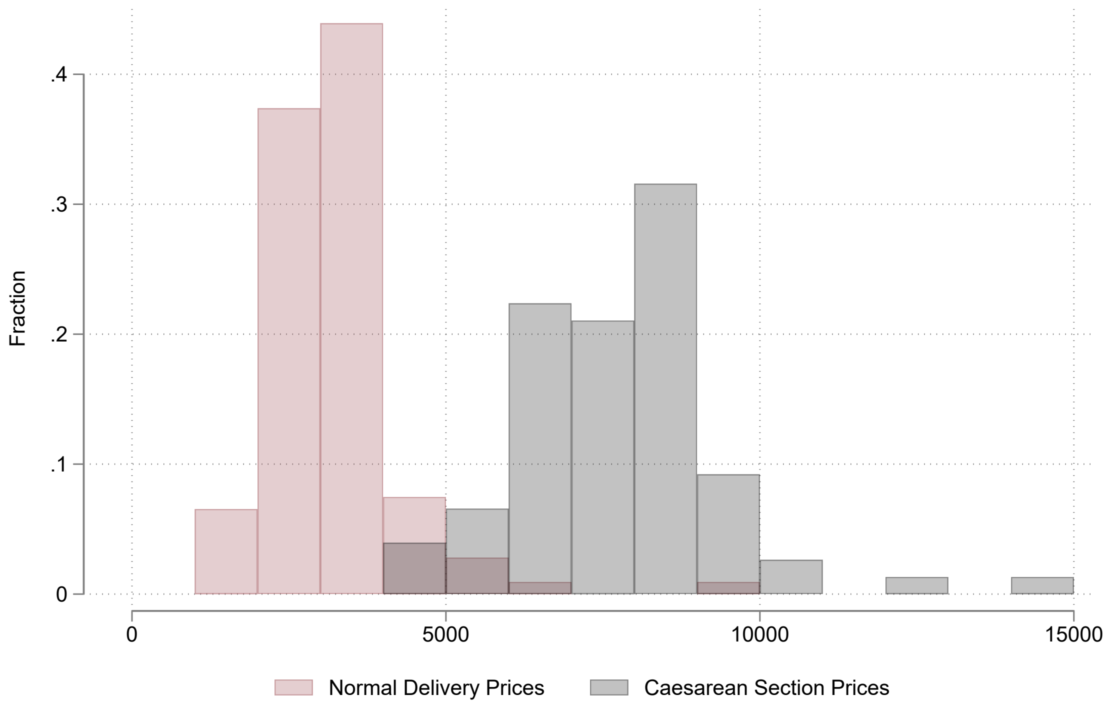
Note: These panels show examples of what hospitals in Malaysia look like based on their categories.

Figure A.5: Determinants of Public and Private Hospital Entry (Full Coefficient Set)



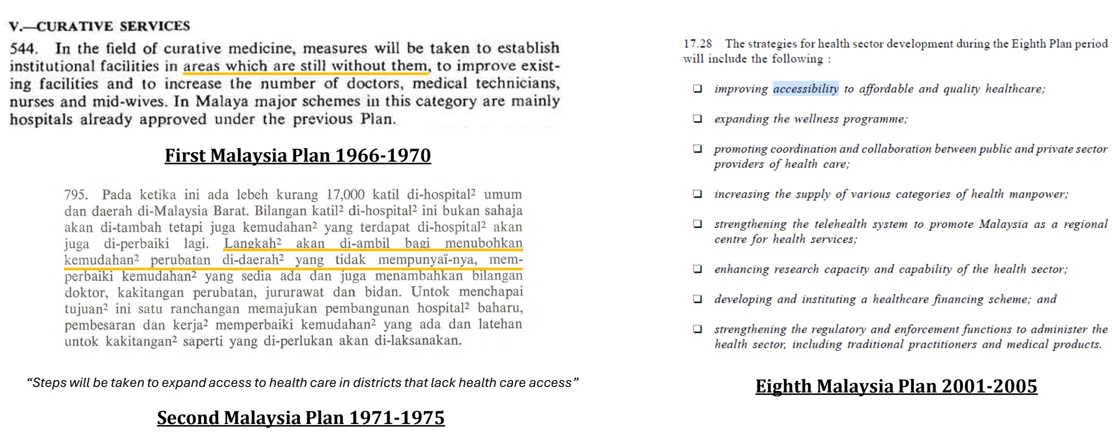
Note: These coefficients are average marginal effects from logit regressions with year fixed effects of public (or private) hospital entry on a set of district characteristics.

Figure A.6: Distribution of Birth Delivery Prices in Private Hospitals (MYR)



Note: This figure shows the distribution of normal (vaginal) delivery and caesarean section prices in private hospitals in Malaysia. Most maternity packages offer only normal delivery packages.

Figure A.7: Selected Excerpts from Malaysia Planning Documents



V.—CURATIVE SERVICES

544. In the field of curative medicine, measures will be taken to establish institutional facilities in areas which are still without them, to improve existing facilities and to increase the number of doctors, medical technicians, nurses and mid-wives. In Malaya major schemes in this category are mainly hospitals already approved under the previous Plan.

First Malaysia Plan 1966-1970

795. Pada ketika ini ada lebeh kurang 17,000 katil di-hospital^2 umum dan daerah di-Malaysia Barat. Bilangan katil^2 di-hospital^2 ini bukan sahaja akan di-tambah tetapi juga kemudahan^2 yang terdapat di-hospital^2 akan juga di-perbaiki lagi. Langkah^2 akan di-ambil bagi menubohkan kemudahan^2 perubatan di-daerah^2 yang tidak mempunyaï-nya, memperbaiki kemudahan^2 yang sedia ada dan juga menambahkan bilangan doktor, kakitangan perubatan, jururawat dan bidan. Untok menchapai tujuan^2 ini satu ranchangan memajukan pembangunan hospital^2 baharu, pembesaran dan kerja^2 memperbaiki kemudahan^2 yang ada dan latehan untok kakitangan^2 saperti yang di-perlukan akan di-laksanakan.

"Steps will be taken to expand access to health care in districts that lack health care access"

Second Malaysia Plan 1971-1975

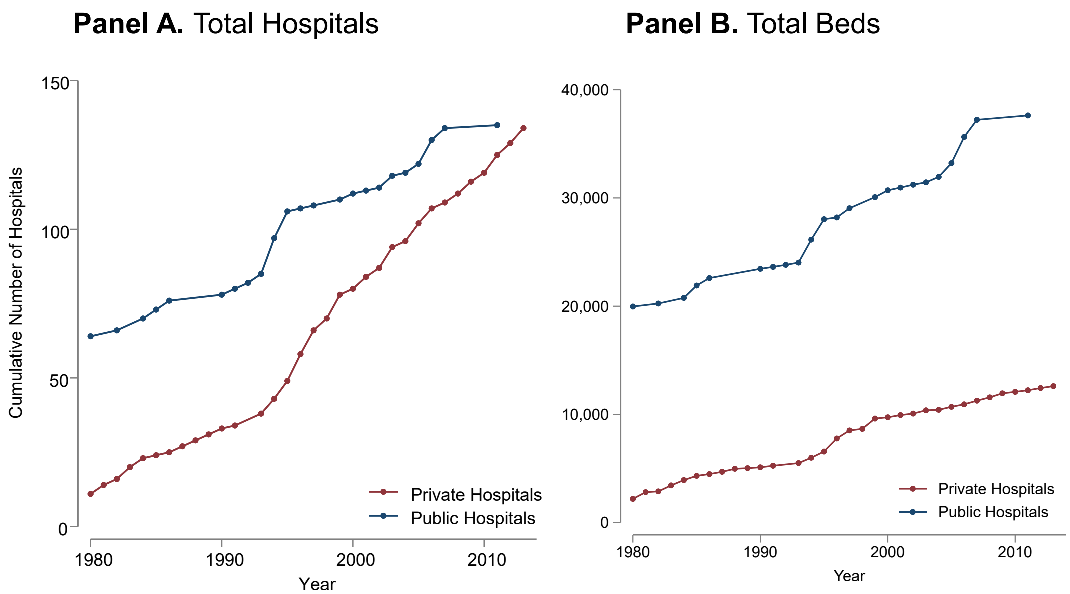
17.28 The strategies for health sector development during the Eighth Plan period will include the following :

- *improving accessibility to affordable and quality healthcare;*
- *expanding the wellness programme;*
- *promoting coordination and collaboration between public and private sector providers of health care;*
- *increasing the supply of various categories of health manpower;*
- *strengthening the telehealth system to promote Malaysia as a regional centre for health services;*
- *enhancing research capacity and capability of the health sector;*
- *developing and instituting a healthcare financing scheme; and*
- *strengthening the regulatory and enforcement functions to administer the health sector, including traditional practitioners and medical products.*

Eighth Malaysia Plan 2001-2005

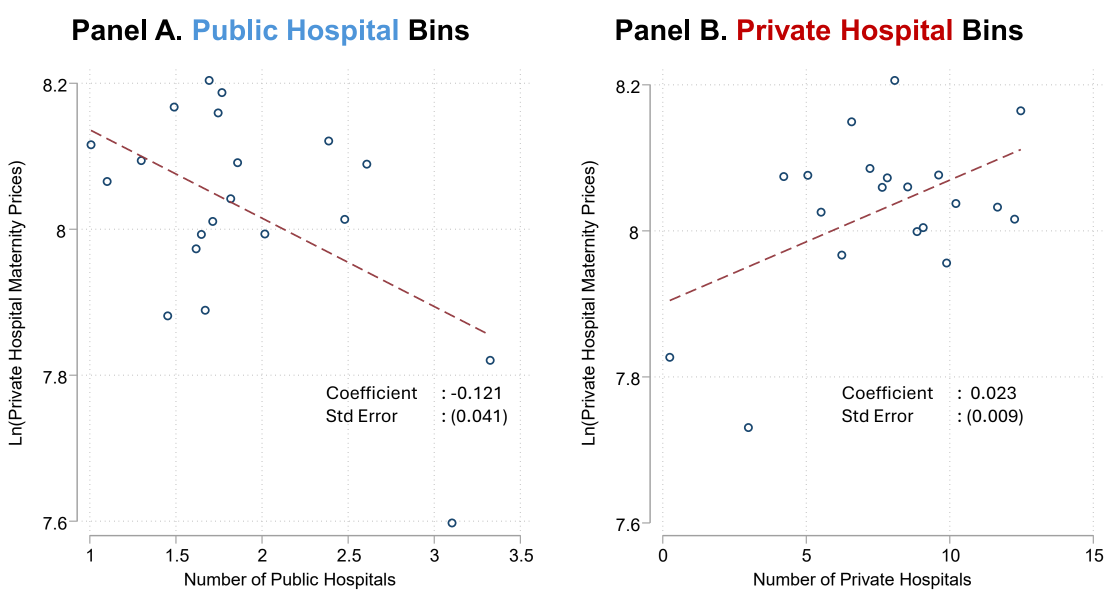
Note: These excerpts are taken from various planning documents related to healthcare development in Malaysia. These panels show the commitment of the Malaysian government in prioritizing access to healthcare.

Figure A.8: Total Count and Beds by Public and Private Hospitals



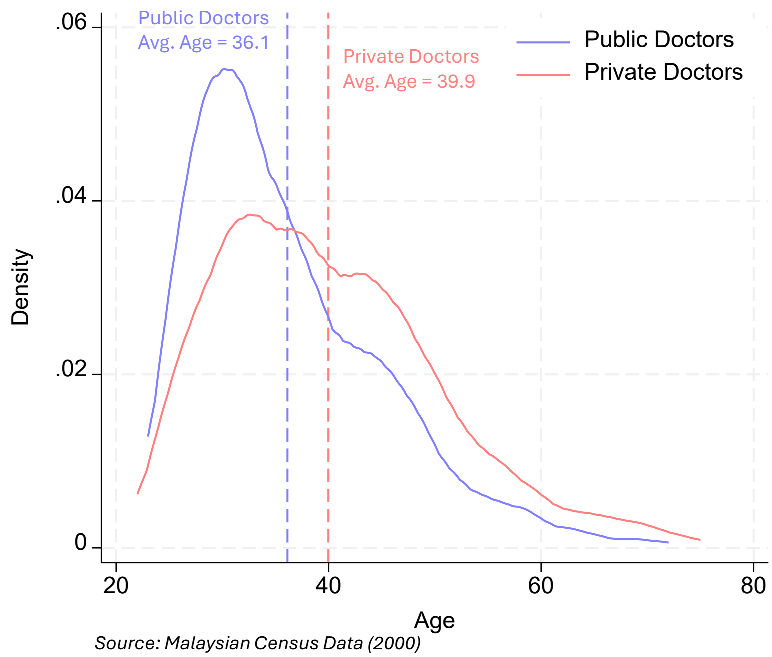
Note: This figure shows the total count of hospitals and the number of beds available in public and private hospitals in Malaysia between 1980 and 2014.

Figure A.9: Private Hospital Normal Delivery Prices by Number of Public/Private Hospitals in District



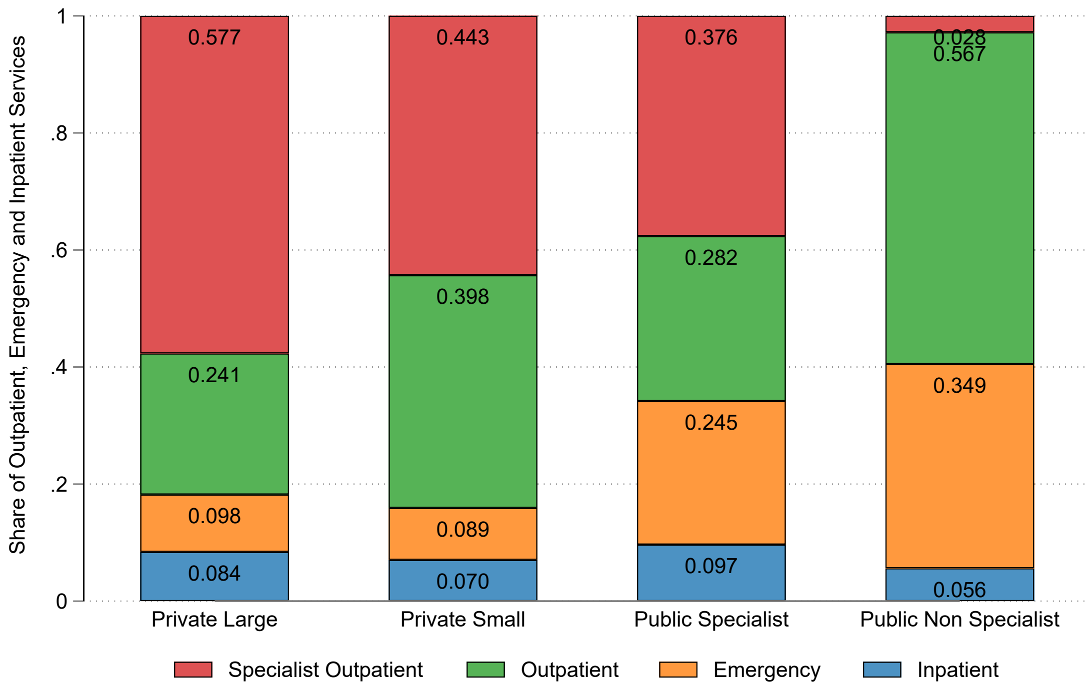
Note: Panel A figure shows a binscatter of private hospital normal delivery prices against the number of public hospitals within the same district. Panel B shows a binscatter against the number of private hospitals within the same district. This figure shows descriptive evidence on public competitive pressures on private pricing.

Figure A.10: Physician Average Age by Public & Private Hospitals



Note: This figure shows the average age of physicians working in public and private hospitals in Malaysia. The difference is approximately 3.8 years. This roughly coincides with the four-year mandatory service obligation in the public sector.

Figure A.11: Proportion of Outpatient, Emergency and Inpatient Visits at Private and Public Hospitals

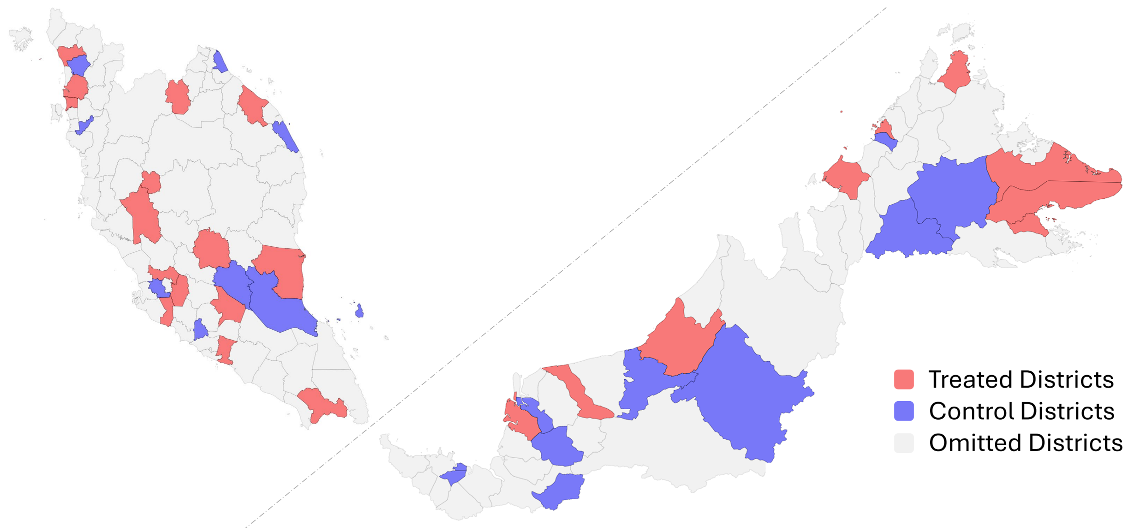


Note: Data from the National Healthcare Establishment Workforce Survey (2013). Panels show the distribution of outpatient, emergency, and inpatient visits across public and private hospitals.

B Further Details on Reduced Form

B.1 Balance and Sample

Figure B.1: Sample of Districts in Event Studies



Note: Red districts are included in the event study design as treated districts, while blue districts are controls. Grey districts are omitted from the event study.

Table B.1: Pre-Treatment Summary Statistics by Treatment Status in 1991

Variable	Treated	Never Treated	Difference	P-value
Panel A. Demographics				
Log Population	11.754 (1.115)	11.057 (1.106)	0.697	0.037
Rural Population Share	0.602 (0.310)	0.782 (0.312)	-0.180	0.063
Average Age	23.717 (2.043)	25.258 (2.536)	-1.541	0.031
Female Share	0.482 (0.024)	0.497 (0.039)	-0.015	0.123
Chinese Share	0.215 (0.151)	0.200 (0.194)	0.015	0.773
Malay Share	0.416 (0.306)	0.447 (0.354)	-0.031	0.752
Indian Share	0.071 (0.080)	0.047 (0.063)	0.024	0.281
Labor Force Participation	0.632 (0.058)	0.642 (0.104)	-0.010	0.669
Panel B. Socioeconomic Status				
Poorest	0.434 (0.260)	0.494 (0.286)	-0.060	0.476
Middle	0.307 (0.129)	0.293 (0.153)	0.014	0.733
Richest	0.258 (0.181)	0.213 (0.205)	0.045	0.446
Panel C. Education				
College/University	0.021 (0.020)	0.021 (0.033)	0.000	0.968
Secondary Completed	0.226 (0.080)	0.222 (0.079)	0.004	0.867
Primary Completed	0.201 (0.041)	0.194 (0.041)	0.007	0.617
Some Primary Education	0.201 (0.030)	0.204 (0.034)	-0.003	0.766
Panel D. Health Facilities				
Dist. to Pub Hosp (km)	36.522 (28.627)	41.423 (40.336)	-4.901	0.630
Dist. to Pri Hosp (km)	102.979 (105.322)	106.691 (112.125)	-3.712	0.907
N Public Hospitals	0.320 (0.557)	0.182 (0.395)	0.138	0.338
N Private Hospitals	0.680 (1.725)	0.273 (0.935)	0.407	0.329
N Specialist Physicians	0.050 (0.111)	0.018 (0.058)	0.032	0.266

Notes: This table compares the 25 treatment districts with the 22 never treated districts based on pre-treatment characteristics in 1991. Standard deviations in parentheses. Unit of observation is districts. All data from the 1991 Malaysian Census and hospital panel data. Distances are straight-line kilometers to facilities from 1km grids, collapsed at the district level. Wealth quintiles are constructed from household assets (electricity, water supply, telephone, automobiles, air conditioning, washing machine, refrigerator, television, VCR, radio, toilet, wall material).

B.2 Full Event Study Coefficients

Table B.2 tabulates the full set of event study coefficients from Equation 7.

Table B.2: Effects of Public Hospital Construction on Private Hospitals (Full Coefficient Table)

Time since Construction	(1) All Public Hospitals	(2) Specialist Public Hospitals	(3) Non-Specialist Public Hospitals
-10	-0.314	-1.190	0.415
	(0.041)	(0.293)	(0.087)
-9	-0.075	-0.527	0.315
	(0.073)	(0.215)	(0.071)
-8	-0.059	-0.329	0.214
	(0.069)	(0.237)	(0.068)
-7	-0.012	-0.310	0.224
	(0.077)	(0.228)	(0.060)
-6	-0.058	-0.286	0.150
	(0.067)	(0.211)	(0.044)
-5	-0.030	-0.166	0.109
	(0.048)	(0.144)	(0.043)
-4	-0.048	-0.130	0.055
	(0.048)	(0.132)	(0.041)
-3	-0.035	-0.090	0.031
	(0.036)	(0.084)	(0.043)
-2	-0.047	-0.077	-0.012
	(0.027)	(0.084)	(0.040)
0	0.082	0.200	-0.070
	(0.048)	(0.095)	(0.034)
1	0.094	0.244	-0.101
	(0.051)	(0.107)	(0.037)
2	0.142	0.355	-0.132
	(0.056)	(0.135)	(0.032)
3	0.127	0.332	-0.134
	(0.058)	(0.140)	(0.032)
4	0.155	0.387	-0.145
	(0.066)	(0.167)	(0.032)
5	0.171	0.436	-0.169
	(0.078)	(0.198)	(0.038)
6	0.344	0.766	-0.198
	(0.123)	(0.261)	(0.039)
7	0.421	0.861	-0.169
	(0.148)	(0.326)	(0.016)
8	0.179	0.392	-0.209
	(0.195)	(0.317)	(0.024)
9	0.213	0.445	-0.261
	(0.223)	(0.345)	(0.030)
10+	0.323	0.673	-0.295
	(0.313)	(0.481)	(0.020)
Observations	846	648	594
R-squared	0.965	0.970	0.946

Standard errors in parentheses, clustered at the district level.

B.3 Alternative Inference Procedures

With 14 specialist treatment events, 11 non-specialist treatment events, and 22 never-treated control districts, conventional clustered standard errors may overstate statistical significance. In this appendix section, I implement two alternative inference procedures that do not rely on asymptotic approximations.

Pairs Cluster Bootstrap. Standard clustered standard errors assume that as the number of clusters grows large, the sampling distribution of the estimator converges to a normal distribution. With as few as 33 clusters in the non-specialist sample, this approximation may be poor. The pairs cluster bootstrap directly estimates this variability by repeatedly resampling the data and computing the statistic each time, rather than relying on the asymptotic formula.

Specifically, I resample entire districts (not individual observations) with replacement ($B = 999$), preserving the within-district correlation structure, and re-estimate the Sun and Abraham (2021) specification on each draw (Cameron et al., 2008). Each bootstrap draw produces a new estimate of the average post-treatment effect. The standard deviation of these 999 estimates is the bootstrap standard error. The two-sided percentile p-value is twice the share of bootstrap estimates that fall on the opposite side of zero from the original coefficient, and the 95 percent confidence interval is constructed from the 2.5th and 97.5th percentiles of the bootstrap distribution.

Table B.3 tabulates the results. For the specialist result, the bootstrap standard error is approximately three and a half times the analytic standard error (0.381 versus 0.108), with a two-sided percentile p-value of 0.042 and a 95 percent confidence interval of [0.013, 1.504]. The specialist result therefore remains statistically distinguishable from zero at the five percent level even under this more conservative procedure. The bootstrap standard error for the non-specialist result inflates by a factor of seventeen (0.161 versus 0.009). The 95 percent percentile confidence interval is $[-0.582, 0.000]$ and the two-sided bootstrap p-value is 0.098, consistent with directional crowd-out that the few-cluster correction does not estimate precisely. The pooled estimate is also statistically indistinguishable from zero under the corrected inference (95 percent CI: $[-0.192, 1.124]$), consistent with the opposing-sign type-specific effects canceling in the pooled average.

Randomization Inference. Randomization inference asks a different question than the bootstrap. Rather than asking whether the standard errors are correctly sized, it asks: could an effect this large have arisen by chance, simply because the specialist hospital events across 14 treated districts happened to be places where private hospitals were growing for reasons unrelated to public hospital construction? The test answers this by asking what the estimated effect would look like if treatment had been randomly assigned to a different set of districts.

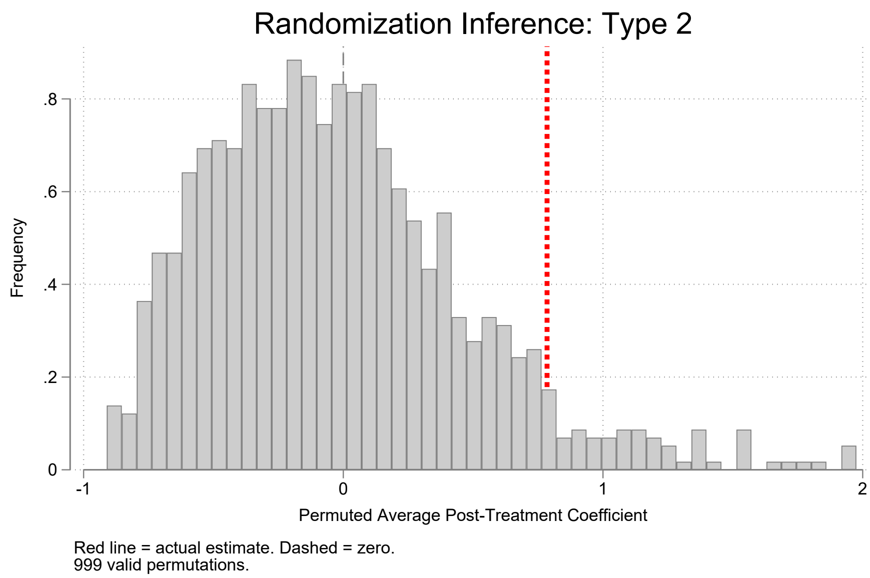
Under the null hypothesis of no treatment effect, I randomly select 14 of the 36 total districts (14 treated plus 22 never-treated), assign them the original vector of treatment years, and designate the remaining 22 as never-treated controls (Young, 2019). I then re-estimate the Sun-Abraham specification and store the average post-treatment coefficient. Repeating this 999 times generates a distribution of placebo effects under the null. If the actual estimate is unusually large relative to this distribution, treatment assignment matters. Figure B.2 shows the result. The permutation distribution is centered near zero, showing the test has power against the null. The actual specialist estimate of 0.785 lies in the right tail, with a one-sided p-value of 0.06 and a two-sided p-value of 0.08.

Table B.3: Alternative Inference for Average Post-Treatment Effects

	All	Specialist	Non-specialist
	(1)	(2)	(3)
Panel A. Main Table Result (Clustered)			
Coefficient	0.465	0.785	−0.171
Standard error	(0.094)	(0.108)	(0.009)
Panel B. Pairs Cluster Bootstrap ($B = 999$)			
Bootstrap SE	0.333	0.381	0.161
SE inflation	3.6×	3.5×	17.1×
Percentile p-value (two-sided)	0.180	0.042	0.098
95% CI	[−0.192, 1.124]	[0.013, 1.504]	[−0.582, 0.000]
Panel C. Randomization Inference ($B = 999$)			
One-sided p-value	—	0.060	—
Two-sided p-value	—	0.078	—

Notes: Panel A reports analytic standard errors clustered at the district level from Table 1. Panel B reports pairs cluster bootstrap results that resample entire districts with replacement. Panel C reports randomization inference p-values from permuting treatment status across all 36 districts (14 treated, 22 never-treated), preserving the original treatment year vector. RI is reported only for the specialist specification (Column 2), which is the paper's main result.

Figure B.2: Randomization Inference Distribution for Specialist Hospital Effects



Notes: Distribution of average post-treatment coefficients from 999 permutations that randomly assign 14 of 36 districts to treatment status. The red vertical line marks the actual estimate (0.785). The dashed line marks zero. The permutation distribution is centered near zero (mean $= -0.026$), showing the test has power against the null.

B.4 Physician Supply: Lag Robustness

The physician supply mechanism occurs with a delay: residency training in Malaysia requires four to six years. The main physician results in Table 3 use lag 3 to capture accumulated effects over a training cycle. Table B.4 reports the effects across lag lengths 0 through 5. For specialist hospitals (Panel A), the effect grows monotonically from 0.547 at lag 0 to 1.081 at lag 3, where it stabilizes. This pattern is consistent with the training timeline: the contemporaneous jump (lag 0) reflects the arrival of senior specialists at the district, while the growth through lags 1–3 captures the first cohorts completing residency training and entering private practice. The stabilization at lag 3 suggests the pipeline reaches a steady state within approximately one training cycle. For non-specialist hospitals (Panel B), the effects are close to zero and statistically insignificant at all lags, showing that the physician supply channel is specific to hospitals with training programs.

Table B.4: Robustness: Effects of Public Hospitals on Private Specialists by Lag Length

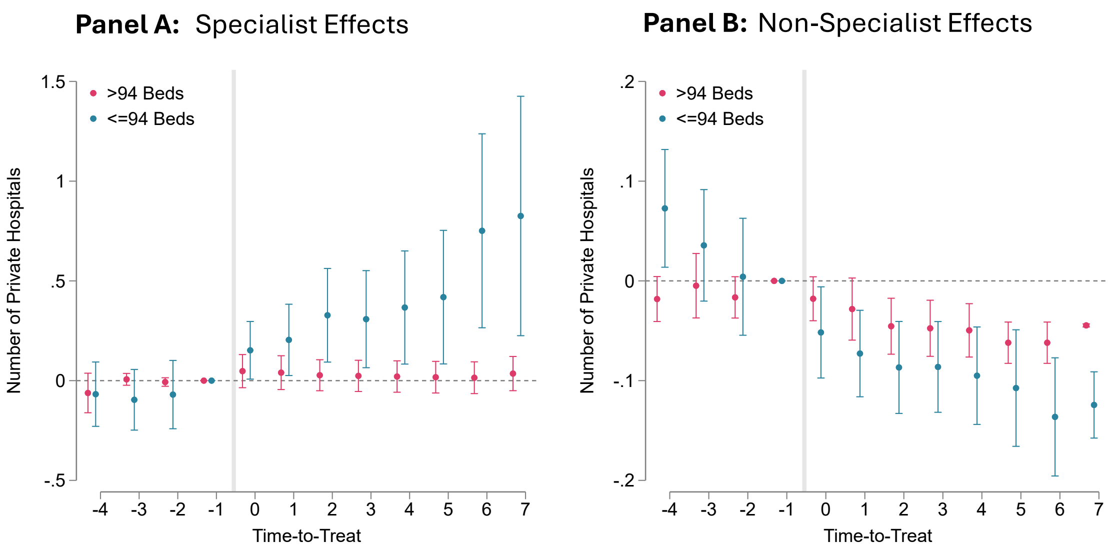
	Private Specialist Physicians (in 100s)					
	Lag 0	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5
Panel A: Specialist Public Hospitals						
Number of Hospitals	0.547	0.772	0.743	1.081	1.081	1.081
	(0.302)	(0.477)	(0.391)	(0.471)	(0.471)	(0.471)
Observations	58	58	60	62	62	62
Mean Dep. Var.	0.309	0.378	0.375	0.411	0.411	0.411
R^2	0.872	0.891	0.891	0.858	0.858	0.858
Panel B: Non-Specialist Public Hospitals						
Number of Hospitals	0.063	0.063	-0.084	-0.084	-0.088	-0.093
	(0.158)	(0.158)	(0.063)	(0.063)	(0.066)	(0.074)
Observations	68	68	66	66	68	68
Mean Dep. Var.	0.159	0.159	0.141	0.141	0.137	0.137
R^2	0.820	0.820	0.880	0.880	0.881	0.881

Notes: Each column presents stacked difference-in-differences estimates with different lag structures. The main physician supply results (Table 3) use lag 3. Effects for specialist hospitals strengthen at longer lags, consistent with residency training programs requiring several years to graduate specialists who then enter private practice. Panel A: specialist public hospitals. Panel B: non-specialist public hospitals. All specifications include district-by-stack and year-by-stack fixed effects. Standard errors clustered at the district level in parentheses.

B.5 Heterogeneous Effects by Hospital Size: Event Study Dynamics

Figure B.3 presents the same event study specification but splitting outcomes to small and large private hospitals. The physician supply mechanism predicts that crowd-in should concentrate among smaller private hospitals, which require fewer specialists to reach minimum viable scale and are more likely to be founded by individual physicians transitioning from public practice. Panel A shows this prediction: small private hospitals (fewer than 94 beds) show a clear positive trend following specialist public hospital construction, with effects growing steadily over the post-treatment window. Large private hospitals show essentially flat effects, consistent with large urban facilities having access to deeper physician labor markets that are less affected by any single public hospital opening. Panel B shows that non-specialist public hospitals produce weakly negative effects for both size categories, with no evidence of differential crowd-out by hospital size. The absence of size heterogeneity in the non-specialist case is consistent with demand competition operating independently of hospital scale, since a new public hospital draws patients regardless of whether the competing private facility is small or large.

Figure B.3: Effects of Specialist and Non-Specialist Public Hospitals on Small and Large Private Hospitals



Note: This figure presents four pre-period and seven post-period event study estimates from Equation 7. Panel A shows the effects of 14 new specialist public hospitals on the number of small (fewer than 94 beds) and large (94 or more beds) private hospitals within the same district. Panel B shows the effects of 11 non-specialist public hospitals. Each dot represents a point estimate with the corresponding 95 percent confidence interval shown as vertical lines. The reference period is $\ell = -1$. Standard errors are clustered at the district level.

B.6 Robustness to Violations of Parallel Trends

I assess the sensitivity of the event study estimates to violations of parallel trends (Rambachan and Roth, 2023). Rather than imposing that parallel trends holds exactly, I bound the post-treatment violation between consecutive years by $\bar{M}$ times the largest pre-treatment violation, the relative magnitudes restriction $\Delta^{RM}(\bar{M})$. I apply the procedure to the estimated coefficients from Table 1.

Table B.5 reports robust confidence sets for the fifteen-year and eight-year post-treatment windows. For the specialist treatment, the averaged effect over fifteen post-treatment years remains distinguishable from zero for violations up to a breakdown value of $\bar{M}^*$ between 0.25 and 0.30. This implies that the estimate survives post-treatment violations up to roughly 30 percent of the largest pre-treatment violation. The pooled effect is more robust, with a breakdown between 0.40 and 0.45, as its pre-treatment coefficients are smaller and more precisely estimated. Overall, moderate violations of parallel trends cannot account for the specialist result, but violations as large as the worst pre-treatment deviation could. This sensitivity analysis should be read in context of other robustness checks.

Table B.5: Sensitivity to Violations of Parallel Trends

Treatment	Window	Estimate	Original CI	$\bar{M} = 0.25$	$\bar{M} = 0.5$	$\bar{M}^*$
Specialist	15 years	0.785	[0.574, 0.995]	[0.101, 1.467]	[−0.523, 2.092]	(0.25, 0.30]
Specialist	8 years	0.448	[0.313, 0.582]	[0.073, 0.821]	[−0.259, 1.155]	(0.30, 0.35]
All public	15 years	0.465	[0.281, 0.648]	[0.144, 0.786]	[−0.077, 1.007]	(0.40, 0.45]
All public	8 years	0.192	[0.133, 0.251]	[0.047, 0.337]	[−0.077, 0.462]	(0.30, 0.35]

Notes: Robust 95% confidence sets under $\Delta^{RM}(\bar{M})$ computed with Rambachan and Roth (2023), using three pre-treatment coefficients. The estimand is the averaged effect over the first 8 or 15 post-treatment years. $\bar{M}^*$ is the breakdown bracket on a grid of step 0.05, reported as the interval between the last grid point at which zero is excluded and the first at which it is included.

B.7 Heterogeneous Effects by Institutional Context: Pre-1996 Corporatization Era

During the early 1990s, the Malaysian government pursued corporatization of public hospitals, proposing to maintain government ownership while operating them as profit-maximizing entities. This policy began with incremental reforms including corporatization of Hospital Kuala Lumpur's cardiac unit in 1992 and contracting out of drug distribution systems in 1994. The policy environment reversed with the 7th Malaysia Plan (1996–2000), when the government recommitted to purely public healthcare provision.

Table B.6 presents event study estimates for the pre-1996 period. All types of public hospitals crowded out private entry during this period, including specialist hospitals. The reversal from uniform crowd-out to heterogeneous effects after 1996 suggests that institutional context matters: crowd-in emerges specifically when public hospitals operate as purely public institutions that train specialists who subsequently enter private practice. During the corporatization era, policy uncertainty about the future structure of public hospitals and the possibility that corporatized facilities would retain trained specialists may have weakened the training pipeline that drives the post-1996 crowd-in result.

Table B.6: Effects of Public Hospital Entry on Private Hospitals: Pre-1996 Period

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	−0.084		
	(0.007)		
Specialist public hospitals		−0.035	
		(0.019)	
Non-specialist public hospitals			−0.095
			(0.009)
Mean Dep. Var.	0.166	0.204	0.181
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
N Districts × Year	1,024	592	784
R^2	0.886	0.943	0.900
Events	42	15	27
Estimator	SA	SA	SA

Notes: Each column presents results from separate regressions for the pre-1996 period when public hospitals faced corporatization pressures. The dependent variable is the number of private hospitals in a district. Standard errors clustered at the district level in parentheses.

B.8 Additional Robustness Tables

This subsection collects robustness checks that address specific threats to the main event study estimates.

Sun-Abraham versus TWFE. A concern with two-way fixed effects estimation under staggered treatment timing is that already-treated units can serve as implicit controls for later-treated units, biasing the estimated treatment effect when effects vary across cohorts (Sun and Abraham, 2021). The Sun-Abraham interaction-weighted estimator addresses this by separately estimating cohort-specific effects and aggregating them. If treatment effect heterogeneity across cohorts is minimal, the two estimators should produce similar results. Table B.7 shows this.

Balancing Regressions. If public hospitals are systematically built in districts that are also experiencing increased private hospital entry, the event study could confound treatment effects with pre-existing trends correlated with district characteristics. To test this, I first predict the number of private hospitals using the district-level covariates from Table B.1, then estimate Equation 7 with this predicted outcome as the dependent variable. Figure B.4 shows that the correlation between public hospital construction and predicted private entry disappears once district fixed effects are included, indicating that the event study design effectively controls for the observable confounders.

Excluding Multiple-Treatment Districts. Three districts received more than one public hospital during the study period. If entry of a second public hospital is endogenous to the private hospital response to the first, including these districts could bias the estimates. Table B.8 drops these three districts. The specialist estimate is somewhat larger in the restricted sample (0.937 versus 0.785), likely reflecting composition as the three excluded districts may have experienced below-average crowd-in. The non-specialist estimate is unchanged (-0.171). Neither change suggests the results are driven by districts with multiple treatments.

Later-Treated as Controls. The main specification uses never-treated districts as the control group. An alternative is to use the later-treated cohort as controls, which ensures that both treatment and control groups eventually receive a public hospital and may therefore be more comparable in terms of unobservable characteristics that predict hospital construction. Table B.9 implements this approach. The specialist estimate of 0.599 is somewhat attenuated relative to the main estimate (0.785) but remains economically large and positive. The non-specialist specification cannot be estimated under this design because both the later-treated control districts and the non-specialist treated districts had zero private hospital entrants.

Comparison Across Staggered DiD Estimators. The recent literature has proposed several alternative estimators for staggered difference-in-differences designs, each making different assumptions about treatment effect dynamics and the appropriate control group. Table B.10 reports the average post-treatment effect from four estimators: Sun-Abraham, Borusyak-Jaravel-Spiess, Callaway-Sant'Anna, and de Chaisemartin-D'Haultfœuille. All four estimators agree on the qualitative pattern: specialist hospitals crowd in and non-specialist hospitals produce null effects. The Callaway-Sant'Anna estimator produces larger point estimates (1.424 for specialist). All four estimators use never-treated districts as controls, so the differences in point estimates reflect differences in how each estimator aggregates cohort-specific effects and handles treatment effect dynamics. Standard errors also vary, reflecting differences in how each accounts for treatment effect heterogeneity and serial correlation.

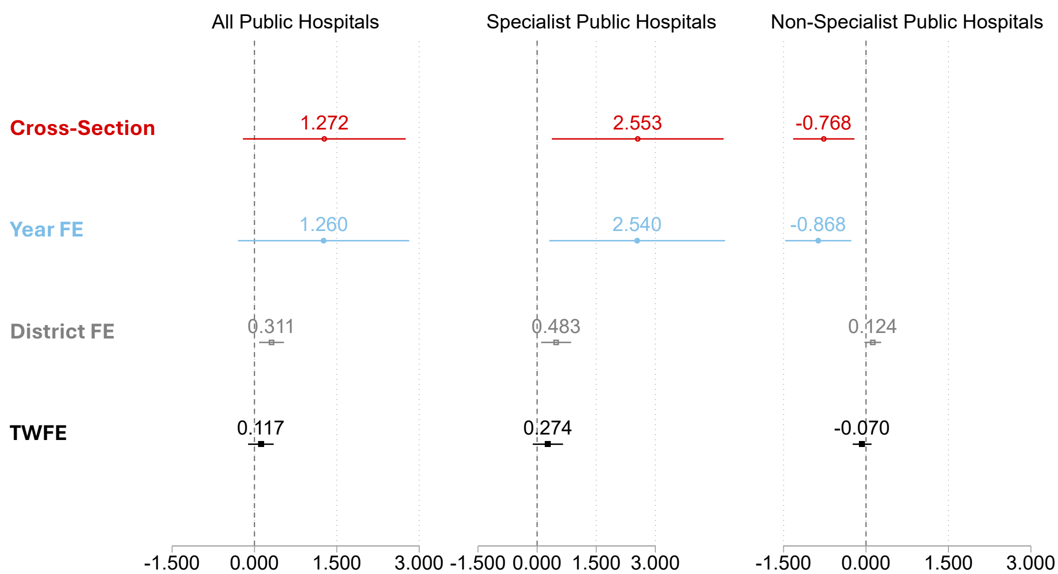
Bed-Size Weighted Effects. If the physician training spillover scales with hospital capacity, larger specialist hospitals should generate proportionally greater crowd-in. Table B.11 tests this by weighting each treatment event by the entering public hospital's bed capacity. The bed-weighted specialist effect (1.069) exceeds the unweighted effect (0.785), consistent with training capacity scaling with hospital size. The non-specialist effect is essentially unchanged under weighting (−0.161 versus −0.171), as expected if crowd-out operates through demand competition rather than a size-dependent mechanism.

Table B.7: Sun-Abraham vs. Two-Way Fixed Effects Estimates

	Number of Private Hospitals		
	All	Specialist	Non-specialist
Panel A. Sun-Abraham			
Post-treatment average	0.465	0.785	−0.171
	(0.094)	(0.108)	(0.009)
Panel B. TWFE with endpoint binning			
Post-treatment average	0.477	0.791	−0.144
	(0.097)	(0.111)	(0.010)
Difference (SA − TWFE)	−0.012	−0.006	−0.027
% difference	2.5%	0.8%	15.8%

Notes: Panel A reproduces the main Sun-Abraham interaction-weighted estimates from Table 1. Panel B reports conventional TWFE estimates with endpoint binning (i.e., binning the earliest and latest relative time indicators to ensure all event times are estimable). The near-identical coefficients for specialist hospitals (less than 1 percent difference) indicate that treatment effect heterogeneity across cohorts is negligible. Standard errors clustered at the district level in parentheses.

Figure B.4: Balancing Regressions



Notes: I first predict the number of private hospitals using district-level characteristics in Table B.1. I then estimate Equation 7 with this predicted outcome as the dependent variable under alternative fixed effects structures. The correlation between public hospital construction and predicted private entry disappears once district fixed effects are included, suggesting that the event study design effectively controls for observable confounders.

Table B.8: Main Effects Robustness: Dropping Multiple Treated Districts

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.457		
	(0.151)		
Specialist public hospitals		0.937	
		(0.114)	
Non-specialist public hospitals			-0.171
			(0.009)
Mean Dep. Var.	1.282	1.709	0.631
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
N Districts $\times$ Year	792	594	594
R^2	0.953	0.957	0.930
Events	22	11	11
Estimator	SA	SA	SA

Notes: Excludes three districts that received multiple public hospitals during 1996–2013. Standard errors clustered at the district level in parentheses.

Table B.9: Main Effects Robustness: Later-Treated as Control

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.684		
	(0.081)		
Specialist public hospitals		0.599	
		(0.088)	
Non-specialist public hospitals			—
Mean Dep. Var.	0.979	1.278	—
District Fixed Effects	Yes	Yes	—
Year Fixed Effects	Yes	Yes	—
R^2	0.980	0.975	—
Events	25	14	—
Estimator	SA	SA	—

Notes: Uses later-treated districts as controls. Column (3) cannot be estimated because both later-treated control districts and non-specialist treated districts had zero private hospital entrants. Standard errors clustered at the district level in parentheses.

Table B.10: Post-Treatment Effects on Private Hospital Entry: Comparison of Estimators

	SA (1)	BJSp (2)	CS (3)	DdH (4)
All public hospitals	0.465	0.296	1.153	0.289
	(0.094)	(0.267)	(0.323)	(0.177)
Specialist public hospitals	0.785	0.659	1.424	0.558
	(0.108)	(0.292)	(0.336)	(0.191)
Non-specialist public hospitals	−0.171	−0.281	−0.208	−0.119
	(0.009)	(0.236)	(0.181)	(0.110)

Notes: Each column presents post-treatment average effects using different estimators for staggered DiD designs. SA: Sun and Abraham (2021). BJSp: Borusyak et al. (2024). CS: Callaway and Sant'Anna (2021). DdH: de Chaisemartin and D'Haultfœuille (2024). All specifications use never-treated districts as controls. Differences in point estimates reflect variation in how each estimator aggregates cohort-specific effects and handles treatment effect dynamics. Standard errors in parentheses.

Table B.11: Bed-Size Weighted Event Study Estimates

	Unweighted			**Bed-Size Weighted**		
	(1)	(2)	(3)	(4)	(5)	(6)
All public hospitals	0.465			1.013		
	(0.094)			(0.109)		
Specialist		0.785			1.069	
		(0.108)			(0.111)	
Non-specialist			−0.171			−0.161
			(0.009)			(0.014)
Mean Dep. Var.	1.303	1.701	0.631	1.303	1.701	0.631
Pre-Treatment Mean	1.440	2.571	0.000	1.440	2.571	0.000
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N Districts $\times$ Year	846	648	594	846	648	594
Events	25	14	11	25	14	11
Estimator	SA	SA	SA	SA	SA	SA

Notes: Columns 1–3 reproduce the main (unweighted) event study results from Table 1. Columns 4–6 weight each treatment event by the entering public hospital's bed capacity. The bed-weighted specialist effect (1.069) exceeds the unweighted effect (0.785), indicating that larger specialist hospitals generate proportionally greater crowd-in, consistent with training capacity scaling with hospital size. The non-specialist effect is essentially unchanged, as expected if crowd-out operates through demand competition rather than a size-dependent mechanism. Standard errors clustered at the district level in parentheses.

B.9 Synthetic Difference-in-Differences

To address concerns about pre-treatment imbalances between treatment and control districts, I re-estimate the main results using the synthetic difference-in-differences estimator (Arkhangelsky et al., 2021). This method constructs synthetic control units that optimally weight both untreated districts and pre-treatment time periods to better match the treated units' characteristics and trends. The specialist SDID estimate (0.692) is positive and similar in magnitude to the main SA estimate (0.785). The non-specialist SDID estimate is attenuated and statistically insignificant, likely reflecting the difficulty of constructing well-matched synthetic controls with only 11 non-specialist treatment events in predominantly rural districts with limited pre-treatment private hospital variation.

Table B.12: Average Treatment Effects on Private Entrants — Synthetic Difference-in-Differences

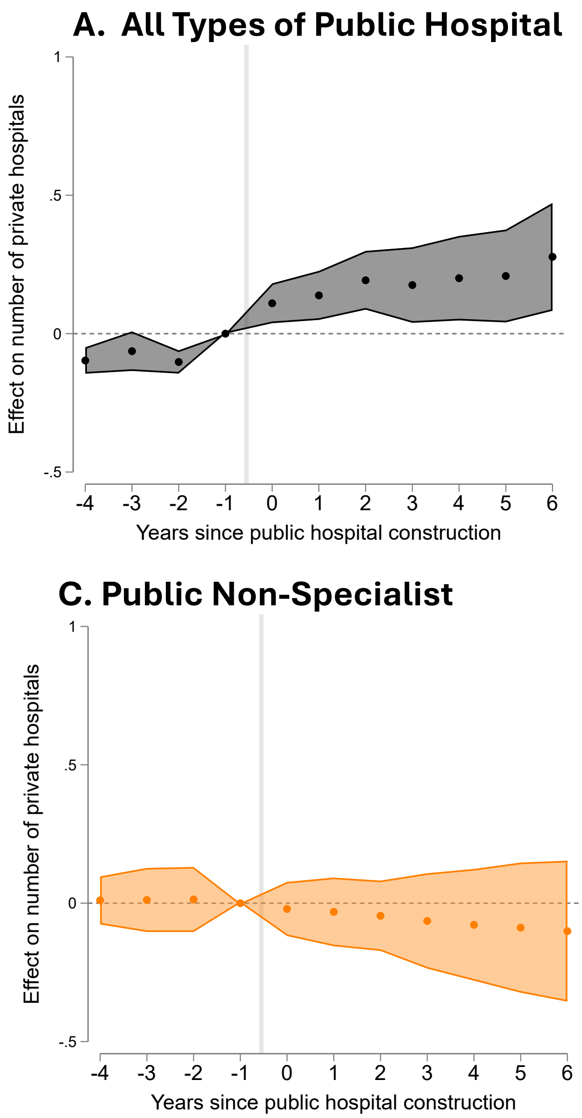
	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.406		
	(0.168)		
Specialist public hospitals		0.692	
		(0.275)	
Non-specialist public hospitals			−0.016
			(0.030)
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Estimator	SDID	SDID	SDID

Notes: Synthetic DiD estimator of Arkhangelsky et al. (2021). Standard errors in parentheses.

B.10 Matching

I use coarsened exact matching (CEM) to balance treatment and control districts on rurality, which shows a significant pre-treatment imbalance, and the number of

Figure B.5: Synthetic Difference-in-Differences: Dynamic Treatment Effects

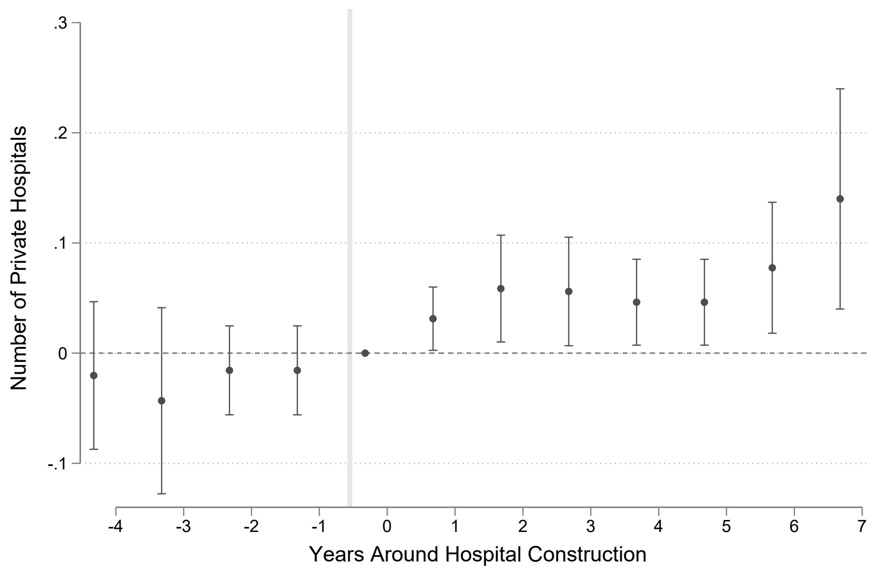


Note: Panel A: all public hospitals. Panel B: specialist public hospitals. Panel C: non-specialist public hospitals. Control units selected using the Synthetic DiD procedure. Shaded area represents the 95 percent confidence interval.

existing public hospitals in 1996, which is a policy-relevant baseline characteristic given the Ministry of Health's stated objective of building in districts without existing public capacity. The matching procedure reduces the sample from 47 districts (25 treatment districts, 22 control) to 28 districts (12 treated, 16 control).

Table B.13 shows that matching removes all statistically significant differences. Figure B.6 presents the event study for the matched sample, pooling specialist and non-specialist public hospitals. The average post-treatment effect is 0.108 additional private hospitals, representing a 31 percent increase relative to the matched sample's pre-treatment mean. The attenuated magnitude relative to the specialist-only estimate (0.785) is expected, since the matched sample includes non-specialist events that reduce the pooled average.

Figure B.6: Event Study of Matching on Private Hospital Entrants



Note: Event study estimates using the Sun and Abraham (2021) estimator on the coarsened exact matched sample (12 treated, 16 control districts). Combines both specialist and non-specialist public hospitals. Matching variables: rurality and number of existing public hospitals in 1996. The average post-treatment effect is 0.108. Standard errors clustered at the district level.

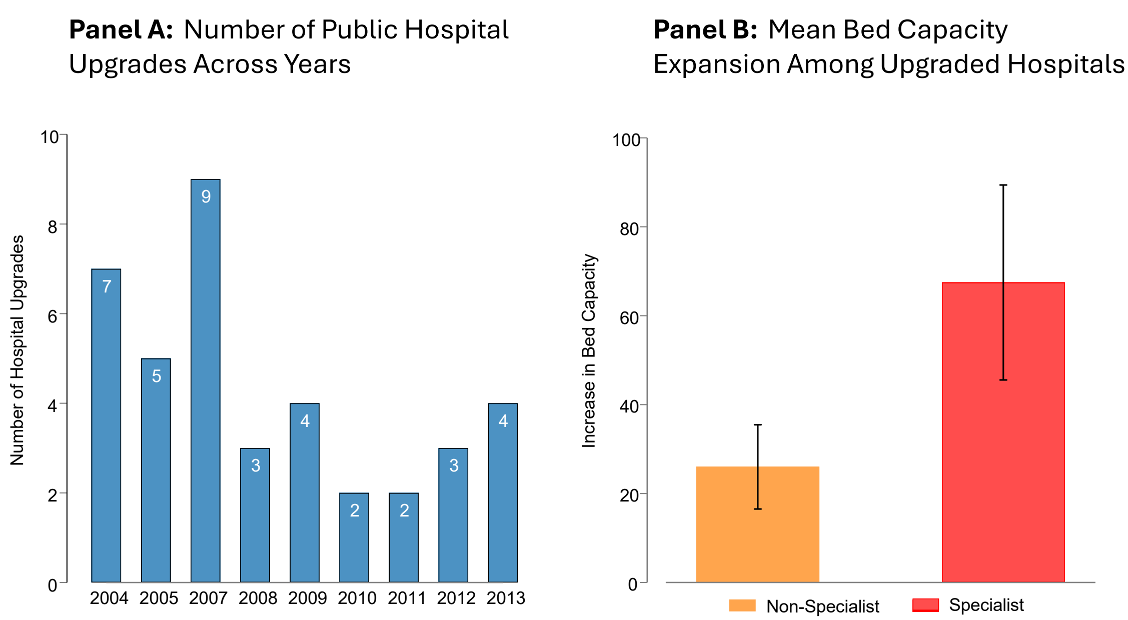
Table B.13: Post-Matching Summary Statistics by Treatment Status

Variable	Treated (N=12)	Never Treated (N=16)	Difference	P-value
Panel A. Demographics				
Log Population	11.186 (0.870)	11.001 (0.869)	0.185	0.582
Rural Population Share	0.843 (0.224)	0.850 (0.262)	-0.007	0.936
Chinese Share	0.127 (0.116)	0.170 (0.198)	-0.043	0.505
Malay Share	0.402 (0.362)	0.431 (0.384)	-0.029	0.839
Indian Share	0.061 (0.077)	0.039 (0.063)	0.022	0.429
Married Share	0.375 (0.015)	0.398 (0.041)	-0.023	0.078
Financial Services Employment Share	0.005 (0.015)	0.006 (0.011)	-0.001	0.894
Panel B. Education				
College/University Education	0.012 (0.014)	0.014 (0.017)	-0.002	0.772
Secondary Education Completed	0.170 (0.077)	0.205 (0.064)	-0.035	0.200
Primary Education Completed	0.188 (0.048)	0.191 (0.038)	-0.003	0.859
Some Primary Education	0.216 (0.028)	0.209 (0.031)	0.007	0.550
Panel C. Age Distribution				
Age <1	0.031 (0.008)	0.026 (0.008)	0.005	0.106
Age 1–4	0.119 (0.019)	0.110 (0.020)	0.009	0.233
Age 5–18	0.340 (0.039)	0.314 (0.044)	0.026	0.119
Age 19–45	0.369 (0.063)	0.388 (0.062)	-0.019	0.451
Age 46–60	0.095 (0.031)	0.101 (0.031)	-0.006	0.612
Age 61–74	0.034 (0.017)	0.047 (0.020)	-0.013	0.077
Age >74	0.012 (0.009)	0.015 (0.008)	-0.003	0.347
Panel D. Health Facilities				
Number of Private Hospitals	0.094 (0.401)	0.375 (1.500)	-0.281	0.534
Number of Public Hospitals	0.125 (0.345)	0.125 (0.342)	0.000	1.000
Number of Private Doctors	12.500 (36.927)	18.750 (62.915)	-6.250	0.762
Distance to Nearest Public Hospital (km)	33.054 (13.585)	36.570 (40.181)	-3.516	0.774
Distance to Nearest	118.454 (88.809)	113.839 (122.953)	4.615	0.913

B.11 Falsification: Hospital Upgrades

To further test the physician training mechanism, I examine an alternative treatment: public hospital upgrades that expand bed capacity without creating new training infrastructure. Unlike new construction, which creates training programs, residency positions, and specialist faculty, upgrades add beds to facilities where teaching programs already exist. If crowd-in operates through new training capacity rather than general service expansion, upgrades should produce muted effects.

Figure B.7: Number of Public Hospital Upgrades by Year



Note: Hospital upgrades data are from the Ministry of Health's annual Health Facts publications, which report bed capacity at each public hospital. Upgrades are defined as year-over-year increases in bed capacity at existing public hospitals, 2003–2012.

Between 2003 and 2012, 35 public hospitals underwent significant bed capacity expansions (Figure B.7). Table B.14 supports the prediction: specialist upgrades produce an effect less than one-third the size of new specialist construction, and the estimate is not distinguishable from zero. Non-specialist upgrades show a negative point estimate similar to new non-specialist construction. The contrast between new construction and upgrades is consistent with the physician training mechanism: creating entirely new training infrastructure generates large physician supply spillovers, while marginally expanding existing capacity does not.

Table B.14: Effects of Public Hospital Upgrades on Private Hospital Entry

	Number of Private Hospitals		
	(1)	(2)	(3)
All upgrades	0.138		
	(0.173)		
Specialist upgrades		0.238	
		(0.215)	
Non-specialist upgrades			−0.170
			(0.169)
Mean Dep. Var.	0.934	1.132	0.631
Pre-Treatment Mean	1.028	1.480	0.000
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	1026	846	594
R^2	0.947	0.946	0.928
Events	35	24	11
Estimator	SA	SA	SA

Notes: Each column presents results from separate regressions examining the impact of public hospital upgrades (2003–2012) using the Sun and Abraham (2021) estimator. Hospital upgrades are defined as year-over-year increases in bed capacity, identified from the Ministry of Health's annual Health Facts publications. Control units are districts that never received upgrades. Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Standard errors clustered at the district level in parentheses.

B.12 Heterogeneous Effects by Private Hospital Size

The physician supply mechanism predicts that crowd-in effects should concentrate among smaller hospitals, which require fewer specialists to reach minimum viable scale. Table B.15 supports this prediction. Specialist hospitals generate large crowd-in effects on small private hospitals (fewer than 94 beds), increasing their count by roughly half relative to the pre-treatment stock, but show essentially no effect on large hospitals (Column 2). This is consistent with small hospitals benefiting most from marginal increases in the local specialist pool, while large hospitals in urban areas face relatively abundant physician supply regardless of public hospital construction. Non-specialist hospitals show directionally negative effects for both size categories (Columns 3–4), consistent with demand competition operating independently of hospital scale. Event study dynamics are in Figure B.3.

Table B.15: Effects on Private Hospitals by Size

	Specialist		**Non-Specialist**	
	Small (1)	Large (2)	Small (3)	Large (4)
Post × Treated	0.727 (0.085)	0.057 (0.029)	−0.173 (0.005)	−0.059 (0.002)
Mean Dep. Var.	0.910	0.790	0.268	0.364
Pre-Treatment Mean	1.429	1.143	0.000	0.000
Observations	648	648	594	594
R^2	0.911	0.982	0.851	0.980
Events	14	14	11	11
District Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Estimator	SA	SA	SA	SA

Notes: Each column presents results from separate regressions using the Sun and Abraham (2021) estimator. The dependent variable is the number of private hospitals in each size category. Small: fewer than 94 beds; large: 94 or more beds. Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Standard errors clustered at the district level in parentheses.

B.13 Further Details on Implied Capacity Calculation

The alternative placement in Section 5.5 involves ranking the 84 candidate districts (no in-window specialist treatment, no pre-1996 specialist hospital) by their predicted capacity to crowd in private hospitals. I rank candidates by predicted private hospital count from a regression of the private hospital count on log population with census-year fixed effects, fit on a stacked panel of three census-year observations (1970, 1980, 1991) per district.

Panel A of Table B.16 reports both sides of the reallocation: the six zero-baseline treated districts whose placements are removed, and the top six candidates that receive them. Predicted crowd-in for each district applies Column (2) of Table 4, $\hat{\tau}_d = \hat{\beta}_1 + \hat{\beta}_2 \left(\ln pop_d - \overline{\ln pop}\right)$, truncated at zero, with log population centered at the estimation-sample mean. Three of the removed districts and two of the added districts (Rompin and Mersing) fall below the population level at which the truncated effect turns positive and contribute zero. Panel B aggregates to the observed and alternative placements. The net gain of 4.6 crowded-in private hospitals equals the additions at the six new sites (5.05) less the forgone crowd-in at the three vacated sites with positive predicted effects (0.42). Beds convert at the 1996–2013 entrant mean of 55 per private hospital.

Table B.16: Implied Private Capacity Under Observed and Alternative Specialist Placement

Panel A: Reallocated placements and predicted crowded-in effects

Removed		Added (top-ranked candidates)	
District	Predicted effect	District	Predicted effect
Temerloh	0.05	Petaling	2.44
Bintulu	0.14	Timur Laut	1.33
Batang Padang	0.23	Larut dan Matang	1.09
Sarikei	0.00	Pasir Mas	0.19
Lahad Datu	0.00	Rompin	0.00
Sepang	0.00	Mersing	0.00
Sum	0.42	Sum	5.05

Panel B: Aggregate implied capacity

	Observed	Alternative	Difference
Crowded-in private hospitals	14.0	18.6	4.6
Crowded-in private beds	770	1,025	255

Notes: Predicted effects apply Column (2) of Table 4 to 1996 log population, truncated at zero. Candidate ranking and the accounting of the alternative column are described in the text. Sums may not match differences exactly due to rounding.